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ON
Smart EV Charging Station with AI-Based Load Prediction
and Management

SUBMITTED IN THE PARTIAL FULFILLMENT OF THE REQUIREMENTS

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IN
ELECTRONICS & TELECOMMUNICATION ENGINEERING

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CERTIFICATE

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DECLARATION BY THE STUDENT(S)

We declare that our the project entitled “**Smart EV Charging Station with AI-Based Load Prediction and Management**” submitted by us for the award of degree Bachelor of Technology in Electronics & Telecommunication Engineering is the record of project work carried out by during the period from **January 2026 to May 2026** under the guidance of **Dr. Rupali B. Patil** and has not formed the basis for the award of any degree, diploma, associate ship, fellowship, titles in this or any other University or other institution of higher learning.

We further declare that the material obtained from other sources has been, duly acknowledged in the project report.

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It gives us great pleasure in presenting “**Smart EV Charging Station with AI-Based Load Prediction and Management**” as our B.Tech. project. Words have never seemed as inadequate as now when we are endeavouring to express our gratitude at the culmination of our B.Tech. Project to all those who have made it possible. Even the best efforts are waste, without the proper guidance and advice of our project guide **Dr. Rupali B. Patil** for the consistent guidance, co-operation, inspiration, practical approach and constructive criticism, which provided us the much-needed impetus to work hard. We sincerely thank to our expert guide **Dr. Sandeep D. Hanwate** for his support and guidance, which played a crucial role in the successful execution of our project & also thanks **Dr. Amol D. Bhoi** Head of E&TC Department for their continuous support & valuable suggestions.

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ABSTRACT

The fast growth of Electric Vehicles (EVs) is putting extra pressure on traditional power systems. EV charging requires high power, and when many vehicles charge at the same time, it can cause transformer overloading, voltage instability, and grid congestion. To solve this issue, this project develops a Smart EV Charging Station with AI-based load prediction and management. The system uses an ESP32 microcontroller as the main controller to manage multiple charging slots. Current and voltage are continuously monitored using ACS712 sensors and voltage sensing circuits to track system performance. The prototype operates on a DC power supply provided by an SMPS along with a battery backup system. When the connected load exceeds the safe limit, a relay-based switching mechanism automatically redistributes the power and activates the battery backup to maintain stable and continuous operation while preventing overload conditions. The system also sends real-time data to an online monitoring dashboard using MQTT, where HiveMQ acts as the platform that transfers the data between the hardware and the dashboard. The dashboard shows real-time monitoring and information such as slot availability, voltage, current, operating status, and billing details. It also allows users to reserve charging slots. A Machine Learning model analyzes previous charging data to predict short-term demand. Based on predicted peak hours, a Dynamic Pricing system adjusts charging rates to encourage off-peak charging. This helps reduce grid stress and improve overall efficiency.

Although the current model is a 12V prototype, the design can be expanded to support 230V AC/DC fast charging systems and smart grid integration. This project combines embedded systems, IoT communication, AI prediction, and intelligent control to create a scalable and practical smart EV charging solution.

Keywords: Smart EV Charging, Machine Learning, Load Prediction, IoT, Dynamic Pricing

TABLE OF CONTENTS

	Page No.
<i>Certificate</i>	<i>i</i>
<i>Declaration by the Student(S)</i>	<i>ii</i>
<i>Acknowledgment</i>	<i>iii</i>
<i>Abstract</i>	<i>iv</i>
<i>Table of Content</i>	<i>v</i>
<i>List of Figures</i>	<i>ix</i>
<i>List of Tables</i>	<i>x</i>
<i>List of Abbreviations</i>	<i>xi</i>

Sr. No.	Title of Chapter	
01	Introduction	1-6
1.1	Overview	2
1.2	Motivation	3
1.3	Problem Definition and Objectives	4
	1.3.1 Problem Statement	4
	1.3.2 Objectives	4
1.4	Project Scope & Limitations	4
	1.4.1 Scope	4
	1.4.2 Limitations	5
1.5	Methodologies of Problem solving	6
02	Literature Survey	7-11
2.1	Market Survey	8
2.2	Literature Survey	9
2.3	Outcomes of Literature Survey	12
03	Smart EV Charging Station: Overview	13-21
3.1	System Architecture	14
	3.1.1 Overview of the System Architecture	16
	3.1.2 Functional Description of Subsystems	16
	3.1.3 Control Logic	17
	3.1.4 Communication and Data Flow	18

3.2	Process Flow	19
04	Smart EV Charging Station: Methodology	22-25
4.1	Methodologies	23
4.1.1	Perception Layer	23
4.1.2	Network Layer	24
4.1.3	Processing Layer	25
4.1.4	Application Layer	25
05	Smart EV Charging Station: System Design	26-44
5.1	System Design	27
5.1.1	Power Source Detection and Prioritization	28
5.1.2	DC Bus Formation and Power Distribution	28
5.1.3	Charging Control and Relay Operation	29
5.1.4	Energy Measurement, Billing, and Performance Estimation	30
5.2	Mathematical Modelling and Performance Calculations	32
5.2.1	Connected Load Calculations	33
5.2.2	System Efficiency of Wired EV Charging System	34
5.2.3	Overall System Efficiency	35
5.3	Hardware Design	37
5.3.1	Vehicle Node Circuit Design	37
5.3.2	Station Controller Circuit Design	38
5.4	Communication Protocol Design	39
5.4.1	MQTT Topic Structure	39
5.4.2	JSON Payload Schemas	40
5.5	Software Design	40
5.5.1	Firmware State Machine	40
5.5.2	Voltage Calibration Algorithm	41
5.5.3	Wi-Fi Manager Captive Portal	41
5.5.4	Neo Pixel LED Status Design	41
5.5.5	Web Dashboard Design	42
5.6	Mathematical Model	43
5.6.1	Voltage Measurement	43
5.6.2	Current Measurement	43

5.6.3	State of Charge Estimation	43
5.6.4	Estimated Charging Time	43
5.6.5	Session Cost Calculation	44
06	Smart EV Charging Station: System Requirements	45-50
6.1	Hardware Requirement	46
6.1.1	Microcontroller Selection Rationale	47
6.2	Software Requirements	47
6.3	Functional Requirements of Proposed System	48
6.3.1	Functional Requirements of Vehicle Node	48
6.3.2	Functional Requirements of Station Controller	48
6.3.3	Functional Requirements of User Portel	49
6.3.4	Functional Requirements of Admin Portel	49
6.4	Non-Functional Requirements	50
07	Smart EV Charging Station: System Implementation	51-56
7.1	Hardware Implementation	52
7.1.1	Vehicle Node Assembly	52
7.1.2	Station Controller Assembly	52
7.2	Firmware Implementation	53
7.2.1	Vehicle Node Firmware Structure	53
7.2.2	Configurable Threshold Macros	53
7.2.3	Charging Detection Algorithm	54
7.2.4	Station Booking Logic	54
7.3	Cloud and Web Dashboard Implementation	54
7.3.1	HiveMQ Cloud Configuration	55
7.3.2	User Portal Implementation	55
7.3.3	Admin Portal Implementation	56
08	Testing and Results	57-67
8.1	Testing Methodology	58
8.1.1	Unit Testing	58
8.1.2	Integration Testing	58
8.1.3	System Testing	58
8.2	Experimental Results	59
8.2.1	MQTT Latency	59

8.2.2	Sensor Calibration Results	59
8.2.3	ACS712 Current Sensor Results	60
8.2.4	Relay Response Time	60
8.2.5	Continuous Operation Test	60
8.2.6	Simulator Fallback Test	61
8.3	Results	61
8.3.1	Hardware Results	61
8.3.2	Software Results	63
09	Conclusions	68-72
9.1	Conclusions	69
9.2	Future Work	70
9.3	Applications	71
	Bibliography	73
	Appendix	75
I	Datasheets	76
II	Copyright Acknowledgment	80
III	Publication	81
IV	Achievements	82
V	Plagiarism Report	85
VI	Sponsorship Letter	87

LIST OF FIGURES

Figure No.	Illustration	Page No.
Figure 1.1	EV Charging Station	2
Figure 3.1	Block Diagram of Smart EV Charging Station	14
Figure 3.2	Flow Chart of Project	20
Figure 5.1	Interactive Block Diagram of Smart EV Charging Station	27
Figure 5.2	Circuit Diagram of Each EV	37
Figure 5.3	Circuit Diagram of Main Station	38
Figure 8.1	Hardware of EV Charging Station	62
Figure 8.2	Slot Booking	64
Figure 8.3	Live Session Panel	64
Figure 8.4	Admin Portal - Slot Management	65
Figure 8.5	Load Analytics and AI Prediction	66
Figure 8.6	Transaction Logs	67
Figure 8.7	Neural Telemetry	67

LIST OF TABLES

Table No.	Illustration	Page No.
Table 5.1	Vehicle Node - PIN Assignment (ESP32-C3)	38
Table 5.2	Station Controller - PIN Assignment (ESP32)	39
Table 5.3	MQTT Topic Structure	39
Table 5.4	Neo Pixel LED Status Colour Code	42
Table 6.1	Hardware Components List	46
Table 6.2	ESP32-C3 vs ESP32 WROOM Comparison	47
Table 6.3	Software Tools and Libraries	47
Table 8.1	MQTT Latency Test Results	59
Table 8.2	Sensor Calibration Results	59
Table 8.3	Relay Switching Response Time	60

LIST OF ABBREVIATIONS

ABBREVIATION	ILLUSTRATION
ADC	Analog-to-Digital Converter
AP	Access Point
BLE	Bluetooth Low Energy
CVRP	Capacitated Vehicle Routing Problem
ESP	Espressif Systems Processor
EV	Electric Vehicle
EVs	Electric Vehicles
GPIO	General Purpose Input/Output
HIVEMQ	Message Queue Telemetry Transport broker by HiveMQ GmbH
HTTP	Hypertext Transfer Protocol
IOT	Internet of Things
IR	Infrared
JSON	JavaScript Object Notation
kWh	Kilowatt-Hour
Li-ion	Lithium-ion
MQTT	Message Queuing Telemetry Transport
NVS	Non-Volatile Storage
PWM	Pulse Width Modulation
QoS	Quality of Service
RGB	Red Green Blue
RMS	Root Mean Square
SoC	State of Charge
SSL	Secure Sockets Layer
TLS	Transport Layer Security
URL	Uniform Resource Locator
VDC	Volts Direct Current
WIFI	Wireless Fidelity (IEEE 802.11)
WS2812B	Addressable RGB LED IC by WorldSemi

Chapter 1

Introduction

INTRODUCTION

1.1 Overview

The transportation sector is undergoing a transformative shift driven by the adoption of electric vehicles (EVs) as a cleaner and more sustainable alternative to internal combustion engine vehicles. Globally, the number of EVs on roads has grown at an exponential rate, with projections indicating over 300 million EVs by 2030. However, this growth is constrained by inadequate charging infrastructure, particularly the absence of smart, remotely monitored, and user-friendly charging stations [4].

This project presents an Smart EV Charging Station with AI-Based Load Prediction and Management that integrates embedded microcontroller intelligence, cloud-based MQTT communication, and web-based user interfaces to create a complete end-to-end charging ecosystem. The system is designed around three ESP32-C3 microcontrollers serving as vehicle nodes and one ESP32 WROOM development board as the central station controller. These devices communicate telemetry data including battery voltage, charging current, power, and state of charge to a cloud MQTT broker in real time [11].



Figure 1.1: Smart EV Charging Station

A key advancement in this system is the integration of Artificial Intelligence (AI)-based load prediction. The system continuously logs historical charging data such as session duration, energy consumption, and slot usage patterns. This data is processed using machine learning techniques (e.g., time-series forecasting) to predict peak demand periods. Based on these predictions, the system can optimize load distribution and

support dynamic pricing strategies, thereby improving energy efficiency and reducing stress on the power source [7].

The project is architected as a multi-layered IoT system: the perception layer consists of current and voltage sensors; the network layer uses Wi-Fi and MQTT over TLS; and the application layer encompasses both a user-facing booking portal and an administrator command center. An intelligent slot booking system enables users to reserve charging slots in 30-minute blocks, view real-time battery status, receive digital receipts, and track live session cost. Administrators gain access to neural telemetry tables, fault diagnostics, load analytics, relay control, and CSV export of transaction records.

A key differentiating feature of this system is the dual-mode operation: when live hardware data is available, it is displayed in real time; when hardware is offline, a built-in software simulator seamlessly mimics charging behaviour, enabling continuous demonstrations. This architecture ensures that client-facing interfaces remain functional under all conditions.

1.2 Motivation

The motivation for this project emerges from three converging technological and social trends: the global rise of EVs, the urgency of accessible charging infrastructure, and the transformative impact of IoT technologies on infrastructure management.

India has set ambitious EV adoption targets under the FAME (Faster Adoption and Manufacturing of Electric Vehicles) policy, aiming to electrify 30% of private cars, 70% of commercial vehicles, 40% of buses, and 80% of two- and three-wheelers by 2030. However, range anxiety the fear of running out of charge remains a primary barrier to EV adoption, and it is directly linked to the scarcity and unreliability of public charging stations [17].

Existing charging infrastructure suffers from critical limitations: users have no way to check real-time slot availability before arriving, there is no mechanism to reserve slots in advance, station operators have limited visibility into utilization and fault conditions, and energy consumption is poorly tracked. Furthermore, most existing stations lack intelligent load management, remote monitoring, and user-accessible booking interfaces [8].

This project addresses these gaps by integrating IoT-based monitoring with AI-driven predictive intelligence, enabling not only real-time visibility but also future demand forecasting. The use of low-cost ESP32 microcontrollers, free-tier cloud MQTT services, and open-source web technologies ensures that the solution is scalable and cost-effective.

1.3 Problem Statement and Objectives

1.3.1 Problem Statement

The rapid growth of Electric Vehicles (EVs) has increased the demand for smart and efficient charging infrastructure. However, many existing EV charging stations face problems such as inefficient power distribution, overload conditions during simultaneous charging, lack of real-time monitoring, and absence of smart slot booking systems. Users often experience long waiting times and limited information about charging slot availability and charging status. In addition, most conventional charging stations do not provide AI-based load prediction, dynamic pricing, or intelligent energy management for handling peak-hour demand efficiently. Therefore, there is a need for a smart IoT-based EV charging system that can manage multiple EVs efficiently while providing real-time monitoring, intelligent load management, slot booking, dynamic pricing, and improved charging station operation.

1.3.2 Objectives

1. To design and develop a smart IoT-based EV charging station using ESP32 microcontrollers for monitoring and controlling multiple charging points.
2. To enable real-time monitoring of voltage, current, and charging status through cloud-based communication using MQTT via the HiveMQ broker.
3. To implement an AI-based load prediction system for forecasting peak demand using historical charging data.
4. To develop a smart web-based platform with slot booking, real-time monitoring, and dynamic pricing for efficient charging station management.
5. To ensure safe and uninterrupted charging through intelligent load balancing and relay-based control mechanisms.

1.4 Project Scope and Limitations

1.4.1 Scope

The proposed Smart EV Charging Station with AI-Based Load Prediction and Management is developed as a low-power prototype to demonstrate intelligent EV charging control, real-time monitoring, and cloud-based supervision. The system operates three 12V DC charging slots and continuously monitors key parameters such as battery voltage, output voltage, and active charging status using ESP32 microcontrollers.

The system implements automatic load balancing based on predefined safety limits. When the load exceeds safe operating conditions, a relay-based control mechanism activates battery backup support to ensure stable and uninterrupted charging operation while preventing overload conditions.

The project also integrates IoT-based remote monitoring with a web-based interface that provides real-time slot availability, booking functionality, and billing simulation. All operational data is logged for analyzing usage patterns and supporting future AI-based load prediction.

Although the current implementation is limited to a 12V/5A prototype, the system architecture is scalable and can be extended for real-world applications such as 230V AC/DC fast charging stations, solar-integrated charging systems, dynamic pricing models, and smart grid communication using MQTT-based connectivity. Thus, the project serves as a foundation for next-generation intelligent EV charging infrastructure.

1.4.2 Limitations

- The system is developed as a 12V low-power prototype, so it does not fully replicate real-world 230V EV fast charging stations in terms of power handling and industrial deployment conditions.
- Sensor readings from components like ACS712 and voltage sensing circuits may have slight inaccuracies due to calibration drift, noise, and temperature variations during continuous operation.
- The system relies on internet connectivity for MQTT-based cloud communication, so any network delay, disconnection, or HiveMQ free-tier restrictions may affect real-time data transmission.
- The billing and payment module is only simulated for demonstration purposes and is not integrated with any real-world payment gateway or financial transaction system.

- The communication range is limited to Wi-Fi coverage only, meaning the system cannot operate in environments requiring cellular or wide-area networking support.
- The central controller does not have a dedicated backup power supply, so in case of a complete power failure, the system may temporarily stop functioning until power is restored.

1.5 Methodology of Problem Solving

The project follows a structured systems engineering methodology, progressing through following phases:

- The proposed Smart EV Charging Station with AI-Based Load Prediction and Management is developed using a systematic IoT and embedded systems approach to solve the problems associated with existing EV charging infrastructure.
- Initially, the problems in existing EV charging stations such as inefficient load management, lack of real-time monitoring, absence of slot booking, and overload conditions were analyzed. Based on these issues, a smart multi-node EV.
- The hardware system uses ESP32-C3 microcontrollers for individual charging nodes and an ESP32 WROOM controller for station management. ACS712 sensors and voltage sensing circuits monitor charging parameters, while relay modules provide automatic load control and battery backup switching during overload conditions for communication and remote monitoring, MQTT protocol is implemented using the HiveMQ cloud broker. The ESP32 devices transmit real-time charging data such as voltage, current, charging status, and slot activity to a web-based dashboard through Wi-Fi connectivity.
- A web-based platform is developed for both users and administrators. The user interface provides features such as slot booking, charging status monitoring, and billing simulation, while the admin interface allows monitoring of system performance and charging activity.
- To improve charging efficiency, historical charging data is collected and analyzed for AI-based load prediction. Based on predicted peak demand conditions, the system supports intelligent load management and future dynamic pricing.
- Finally, the complete system is tested under different charging and load conditions to verify sensor performance, MQTT communication, relay operation, and overall system reliability.

Chapter 2

Literature Survey

LITERATURE SURVEY

2.1 Market Survey

The rapid growth of Electric Vehicles (EVs) has significantly increased the demand for efficient and intelligent EV charging infrastructure. With rising environmental concerns, government policies promoting clean energy, and the shift toward sustainable transportation, EV adoption is increasing globally as well as in India. This growth has created new challenges in charging station management, including load balancing, energy optimization, scheduling, and grid stability.

The global EV charging market has witnessed substantial expansion over the past decade. Advances in smart grid technology, Artificial Intelligence (AI), Internet of Things (IoT), and cloud computing have enabled the development of intelligent charging systems capable of real-time monitoring, predictive analysis, and automated control. Leading companies such as Tesla Supercharger, ChargePoint, ABB, Siemens, and Tata Power EV Charging have introduced advanced charging solutions with features like fast charging, smart energy management, and remote monitoring.

Despite these advancements, existing EV charging systems face several limitations such as uneven load distribution, peak demand stress on power grids, lack of real-time optimization, and limited integration of predictive analytics. Many systems rely on static scheduling and do not adapt dynamically to changing demand conditions. Additionally, high infrastructure cost and complexity limit widespread deployment, especially in developing regions.

In India, where EV adoption is rapidly increasing, efficient charging infrastructure is still evolving. Urban areas face issues such as peak load demand, limited charging stations, and inefficient energy utilization. Affordable, scalable, and intelligent charging solutions are highly required to support this growing ecosystem.

This scenario presents a strong opportunity for innovation in smart EV charging systems. Our proposed system aims to address these challenges by integrating AI-based load prediction, IoT-based real-time monitoring, and intelligent energy management. By ensuring efficient power distribution, reducing peak load stress, and improving charging efficiency, the system aims to enhance overall performance and reliability.

The proposed system focuses on developing a cost-effective, scalable, and intelligent EV charging solution suitable for real-world deployment. By combining predictive analytics, real-time data processing, and smart scheduling, it aims to set new standards in EV charging infrastructure and contribute toward sustainable energy management.

2.2 Literature Survey

The rapid growth of electric vehicles (EVs) has introduced several challenges in efficient charging station management, including load balancing, scheduling, energy optimization, and grid stability. To address these issues, researchers have proposed various solutions based on Artificial Intelligence (AI), Internet of Things (IoT), and blockchain technologies.

Khan et al., 2024 [1] proposed an AI-driven dynamic power allocation framework for EV charging stations. Their model intelligently distributes available power among multiple connected EVs to improve charging efficiency and maximize station utilization. Simulation results showed significant performance improvements; however, the system suffers from high computational complexity and lacks validation in large-scale real-time environments.

Hossen et al., 2024 [2] introduced an AI-enabled framework based on the Open Charge Point Protocol (OCPP) for secure and smart EV charging. The study focused on improving communication security and evaluated the system under real-world conditions. While the framework demonstrated practical feasibility, it requires advanced communication infrastructure and has limited scalability validation.

Ren et al., 2023 [3] developed a robust EV charging scheduling strategy considering uncertainties in electricity pricing and fluctuating charging demand. Their approach effectively optimized economic performance under uncertain conditions. However, the work mainly focuses on theoretical modeling and does not include hardware-level implementation.

Saldarini et al., 2023 [4] analyzed smart charging strategies using traffic flow and EV charging simulations on highways. The study demonstrated effective peak load reduction and improved energy distribution. Despite promising results, the approach was only validated through simulations and lacks real-time system implementation.

Zhao et al., 2024 [5] proposed an AI-based load forecasting model using IoT-enabled wireless sensors. Their system improved prediction accuracy for EV charging demand and enhanced energy management efficiency. However, it requires large datasets and significant computational resources for training and deployment.

Zhang and Cai, 2018 [6] presented a dynamic EV charging scheduling model integrating photovoltaic (PV) systems with grid power. The system successfully reduced peak grid load and increased renewable energy utilization. Nevertheless, its performance heavily depends on solar energy availability and accurate prediction of EV arrivals.

Chen and Folly, 2023 [7] applied AI techniques for optimized EV charging and discharging scheduling combined with dynamic pricing strategies. Their results showed improvements in both economic efficiency and operational performance. However, the study is primarily simulation-based and lacks hardware implementation.

Dasi et al., 2024 [8] proposed an IoT-based intelligent energy management system for EV charging stations. The system enables real-time monitoring, cloud-based control, and efficient energy utilization. Although effective, the architecture increases system complexity and deployment costs.

Koresh et al., 2024 [9] developed an IoT-based EV charging slot reservation system that allows users to book charging slots in real time. The system improves user convenience and station utilization. However, it does not incorporate AI-based load prediction or dynamic pricing mechanisms.

Singh et al., 2024 [10] proposed an AI-integrated blockchain framework for secure EV charging networks. The system enhances data security, transparency, and trust among users. Despite these advantages, the framework introduces high computational overhead and increased architectural complexity.

Nizam et al., 2024 [11] proposed a real-time energy monitoring system for renewable EV charging stations using an ESP32 integrated with Modbus, MQTT, and ESP-NOW communication protocols. The system enabled efficient real-time monitoring and communication for EV charging infrastructure. Experimental results demonstrated reliable data transmission and energy management performance. However, the work mainly focused on monitoring architecture and did not make AI-based load prediction.

Tucker et al., 2022 [12] developed a real-time smart EV charging framework for workplace charging stations using predictive scheduling algorithms and charging demand analysis. The proposed system minimized electricity cost and reduced peak demand stress while ensuring adequate charging for EV users. However, the system required complex forecasting models and large-scale deployment infrastructure.

Li et al., 2022 [13] proposed a reinforcement learning-assisted deep learning approach for EV charging power forecasting. The system used LSTM and reinforcement learning techniques to predict charging demand under uncertain conditions. Results showed improved forecasting accuracy and efficient load management. Nevertheless, the model required large datasets and high computational resources for training.

Bhosale et al., 2025 [14] designed an ESP32-based smart EV charging station monitoring system with IoT integration for real-time measurement of voltage, current, temperature, and charging status. The system improved monitoring efficiency and provided protection against abnormal charging conditions. However, the work did not include AI-based prediction or dynamic pricing mechanisms.

Gomes et al., 2026 [15] developed a low-cost ESP32-based smart lithium-ion battery charging and monitoring system integrated with MQTT-based IoT communication. The system supported real-time monitoring of charging parameters and cloud-based data logging. Although effective for battery monitoring applications, the architecture was limited to low-power charging systems and lacked intelligent load management features.

Dzahir and Chia, 2023 [16] evaluated ESP32 performance for MQTT-based real-time IoT monitoring systems. The study showed that ESP32 supports efficient low-power communication for continuous sensor monitoring. However, the work mainly focused on communication efficiency rather than EV charging applications.

Lee et al., 2020 [17] presented an Adaptive Charging Network (ACN) framework for smart EV charging stations with real-time monitoring and adaptive scheduling algorithms. The proposed system improved charging efficiency and reduced congestion in heavily loaded charging environments. However, the framework requires complex infrastructure and large-scale deployment support.

2.3 Outcomes Identified from Literature Survey

From the detailed analysis of existing research papers related to EV charging systems, AI-based energy management, IoT monitoring, and smart scheduling techniques, the following outcomes were identified:

1. Existing EV charging systems mainly focus on smart monitoring, scheduling, and energy optimization using AI and IoT technologies.
2. AI-based load prediction techniques improve charging efficiency, reduce peak load stress, and enhance power management in charging stations.
3. IoT-enabled systems provide real-time monitoring of voltage, current, battery status, and charging conditions through cloud communication.
4. Dynamic charging scheduling and adaptive power allocation help in balancing energy demand among multiple EVs.
5. Integration of renewable energy sources and smart grid technologies improves sustainable energy utilization in EV infrastructure.
6. MQTT, OCPP, and ESP32-based communication systems enable reliable real-time data transmission and remote monitoring.
7. Most existing works are simulation-based and lack low-cost real-time hardware implementation with integrated AI prediction models.
8. Existing systems often suffer from high computational complexity, scalability issues, and dependency on large datasets.
9. There is a need for a cost-effective, real-time smart EV charging station that combines AI-based load prediction, IoT monitoring, and intelligent energy management in a single integrated system.

Chapter 3

Smart EV Charging Station: Overview

SMART EV CHARGING STATION: OVERVIEW

3.1 System Architecture

The system architecture of the proposed Smart EV Charging Station with AI-Based Load Prediction and Management is designed to provide an efficient, scalable, and intelligent charging solution by integrating embedded hardware, real-time sensing, wireless communication, and cloud-based monitoring.

The architecture follows a layered and modular design, ensuring flexibility, reliability, and ease of expansion. It enables seamless coordination between physical hardware components, control logic, communication protocols, and user-level applications.

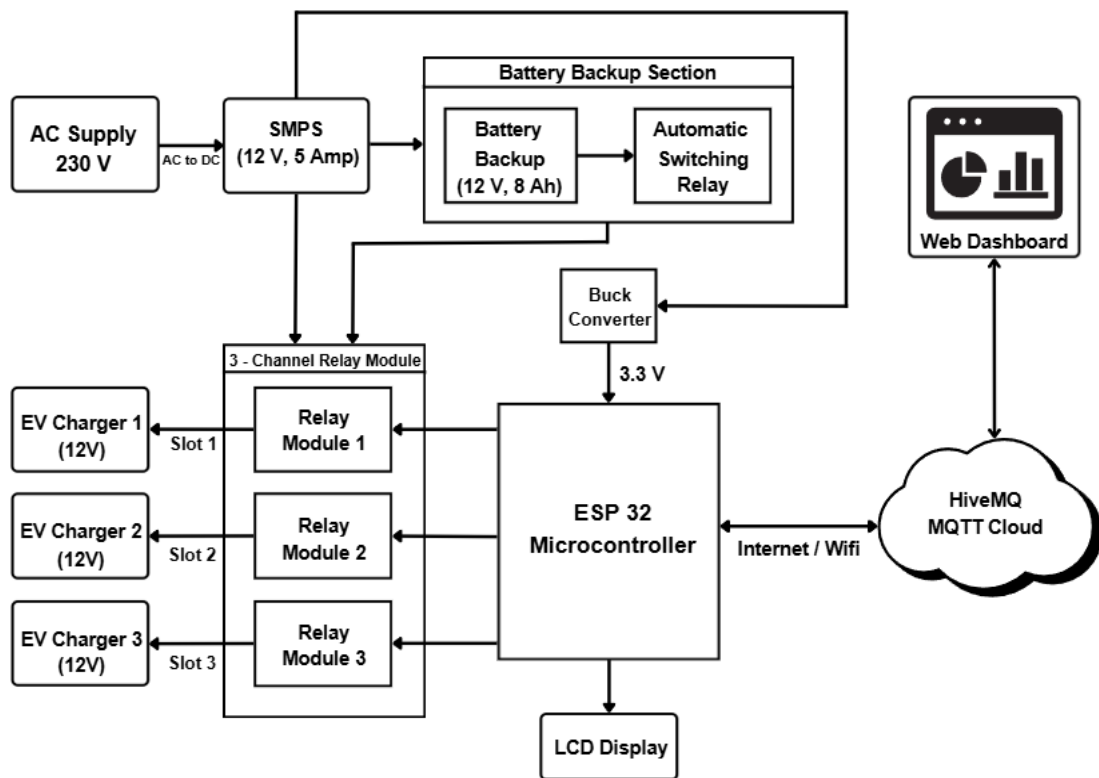


Figure 3.1: Block Diagram of Smart EV Charging Station

The system architecture of the Smart EV Charging Station is built around a modular and layered design, making it scalable, reliable, and straightforward to expand. As illustrated in Figure 3.1, the architecture integrates embedded hardware, wireless communication, real-time control, and cloud-based monitoring into one cohesive framework.

1) Power Supply Section

The system draws power from a 230V AC mains supply, which is fed into a Switched

Mode Power Supply (SMPS) that converts it to a regulated 12V DC at 5A. This serves as the primary power source for the entire station. A buck converter further steps this down to 3.3V to power the ESP32 microcontroller safely.

2) Battery Backup Section

A 12V 8Ah battery backup unit is connected alongside the SMPS through an automatic switching relay. Whenever the main supply is interrupted or the load exceeds the SMPS capacity, this relay activates and seamlessly transfers power responsibility to the battery, ensuring uninterrupted station operation.

3) EV Charging Load Section

Three EV chargers, each operating at 12V are connected to the system as Slot 1, Slot 2, and Slot 3. Each slot represents an independent charging point capable of serving a different vehicle with varying power demands. The slots are physically separated and independently controlled, allowing the station to serve multiple users simultaneously.

4) 4-Channel Relay Module

Each charging slot is connected through a dedicated relay module — Relay Module 1, 2, and 3 respectively. These relays act as electronically controlled switches, allowing the ESP32 to activate or deactivate any slot individually based on booking status, load conditions, or safety triggers. The relay modules provide essential electrical isolation between the low-voltage control circuitry and the higher-current charging lines.

5) Main Controller - ESP32 Microcontroller

The ESP32 sits at the center of the architecture and serves as the brain of the entire system. It receives sensor inputs, executes control logic, drives the relay modules, displays status on the LCD, and communicates with the cloud over Wi-Fi. Its dual role as both a local controller and an IoT gateway makes it the most critical component in the design.

6) Communication - HiveMQ MQTT Cloud

The ESP32 connects to the internet via Wi-Fi and communicates with the HiveMQ MQTT Cloud broker. Sensor data such as voltage, current, power, and slot status is continuously published to structured MQTT topics. At the same time, the system subscribes to command topics to receive booking requests and control instructions from the web dashboard.

7) Web Dashboard

The web dashboard serves as the user-facing interface for the entire system. It subscribes to MQTT topics via WebSock and displays live real-time data, slot

availability, charging session details, and system health, all updated in real time. Administrators can also send commands such as slot booking, force relay control, and session termination directly from the dashboard.

Overall, the architecture establishes a closed-loop intelligent control system where power management, sensing, processing, cloud communication, and user interaction are tightly integrated - working together to deliver a smart, efficient, and reliable EV charging solution.

3.1.1 Overview of the System Architecture

The Smart EV Charging Station is built on a modular, multi-node architecture that connects embedded hardware, IoT communication, and AI-based decision-making into a single cohesive system. It is structured around three vehicle nodes and one central station controller - each node monitors its own battery parameters, while the controller handles slot allocation, relay switching, and overall coordination.

The system operates across two core domains:

1) Power Flow Domain

A 12V 5A SMPS serves as the primary supply, backed by a 12V battery unit for uninterrupted operation during failures. Power reaches each charging slot through relay modules controlled by the ESP32, with DC motors simulating real EV load conditions.

2) Control and Data Flow Domain

Sensors continuously capture voltage and current data, which the ESP32 processes and transmits to the cloud via MQTT. The controller also handles vehicle detection through IR sensors and executes relay logic based on bookings and system state.

Beyond real-time control, the system includes an AI-based load prediction module that analyzes historical data to forecast peak demand periods and enable intelligent load management, dynamic pricing, and overload prevention. A web-based portal gives users and admins the ability to book slots, monitor charging sessions.

3.1.2. Functional Description of Subsystems

The proposed Smart EV Charging Station system is divided into multiple functional subsystems, each responsible for a specific operation. These subsystems work together to ensure efficient charging, real-time monitoring, intelligent control, and predictive load management.

1. Power Management and Charging Control Unit

This unit handles power delivery to all charging slots. The ESP32 monitors total load via the ACS712 sensor and controls a 4-channel relay module to activate or deactivate individual slots based on booking status and safety conditions. If the load exceeds safe limits or the main supply fails, the system automatically switches to battery backup, ensuring continuous and stable operation.

2. Vehicle Node Monitoring Unit

Each of the three vehicle nodes uses an ESP32-C3 microcontroller paired with an ACS712 current sensor and a voltage divider circuit. Battery packs of different capacities (5500 mAh, 3300 mAh, 2200 mAh) along with DC motor loads simulate varied EV conditions. Each node calculates voltage, current, power, and State of Charge (SOC) in real time and transmits this data to the cloud via MQTT.

3. IoT Communication and Data Acquisition Unit

All ESP32 devices connect to Wi-Fi and communicate with HiveMQ Cloud using MQTT over TLS. Data is structured in JSON format and published to dedicated topics. The system supports bidirectional communication, devices send telemetry upward while the application sends booking commands downward. All data is stored in a cloud database for future analysis and AI model training.

4. AI-Based Load Prediction and Analytics Unit

Historical charging data including slot usage, power consumption, and session duration is used to train machine learning models for time-series forecasting. The AI engine predicts future load demand, peak usage hours, and slot occupancy trends. Based on these predictions, the system can pre-optimize power distribution, prevent overloads, and apply dynamic pricing, shifting the station from a reactive device to a proactive energy management system.

5. User Interface and Management Unit

The web application includes two portals. The User Portal allows slot booking, real-time status viewing, and session monitoring. The Admin Portal provides full control over all nodes, live telemetry, manual relay control, and access to usage analytics and fault logs. Both portals communicate with the system via MQTT for live updates.

3.1.3 Control Logic

The control logic of the proposed Smart EV Charging Station is designed to ensure safe, efficient, and uninterrupted charging operation under varying load conditions.

The control logic follows a step-by-step sequence to ensure safe and efficient operation:

- **Initialization:** On startup, the ESP32 initializes all sensors, relays, and communication modules. All relays remain OFF until a valid booking is received.
- **Slot Activation:** When a user books a slot through the web portal, a command is sent via MQTT. The ESP32 validates and activates the corresponding relay, starting the charging session with live monitoring.
- **Load Monitoring:** The ACS712 sensor tracks total current draw continuously. If load approaches the safe threshold, the system limits active slots or pauses lower-priority sessions.
- **Backup Switching:** On main supply failure or overload, the system shifts to battery backup automatically and returns to SMPS once conditions normalize.
- **Charging Termination:** Charging stops when the booking expires, the vehicle is removed, the battery is full, or a fault condition is detected.
- **AI-Assisted Optimization:** Predicted peak hours allow the system to proactively limit slot availability and schedule relay activation, reducing overload risk before it occurs.

This combination of real-time reactive control and AI-based predictive logic ensures robust and intelligent power management.

3.1.4 Communication and Data Flow

The communication and data flow of the system follow a structured IoT-based architecture that enables seamless interaction between hardware components, cloud infrastructure, and user interfaces.

The data flow follows a structured, layered IoT pipeline:

- **Perception Layer:** Vehicle nodes and the station controller continuously capture voltage, current, and load data through sensors.
- **Controller Layer:** ESP32 units process raw sensor readings locally, compute key parameters like SOC, and package the data into JSON payloads.
- **Network Layer:** Data is published to MQTT topics via Wi-Fi and TLS-secured connection to HiveMQ Cloud. The system also subscribes to command topics for receiving booking and control instructions.
- **Cloud and Storage Layer:** HiveMQ acts as the central broker. All incoming data is stored in a database, building a historical record used for performance analysis and AI model training.

- **Application Layer:** Both user and admin portals subscribe to MQTT topics via WebSockets, displaying real-time data and sending control commands back to the station controller.
- **AI Integration:** Historical and live data feeds into the prediction model, whose outputs guide slot allocation, load scheduling, and system behavior during peak demand periods.

3.2 Process Flow

The process flow of the Smart EV Charging Station covers the complete sequence of operations from system startup to charging completion as illustrated in Figure 3.2. The entire flow is managed by the ESP32 microcontroller, which handles power decisions, slot control, real-time monitoring, and cloud communication in a continuous loop. This flowchart represents the working process of a Smart EV Charging System with Battery Backup and IoT Monitoring.

The process starts when 230V AC supply is provided to the system. The SMPS converts AC power into 12V DC, which is then used for system operation. The system first checks whether AC power is available. If yes, charging operates using SMPS power; if no, the 12V battery backup is activated through a relay to ensure uninterrupted operation.

Next, power is supplied to the ESP32 controller through a buck converter, which reduces voltage to 3.3V. The ESP32 continuously checks the EV charging slots to detect whether any electric vehicle is connected.

- If no EV is connected, the system remains in monitoring mode.
- If an EV is connected, the corresponding relay is activated to start charging.

The system then checks the load condition:

- Under normal load, charging continues using AC supply.
- During high load or multiple EV connections, the battery backup is enabled to support charging and maintain stability.

During charging, the system continuously monitors voltage, current, and charging status. This data is sent to the cloud using WiFi and HiveMQ MQTT protocol, and the web dashboard is updated in real time.

Once charging is completed, the relay turns OFF, final charging status is uploaded to the cloud, the dashboard is updated, and the process ends.

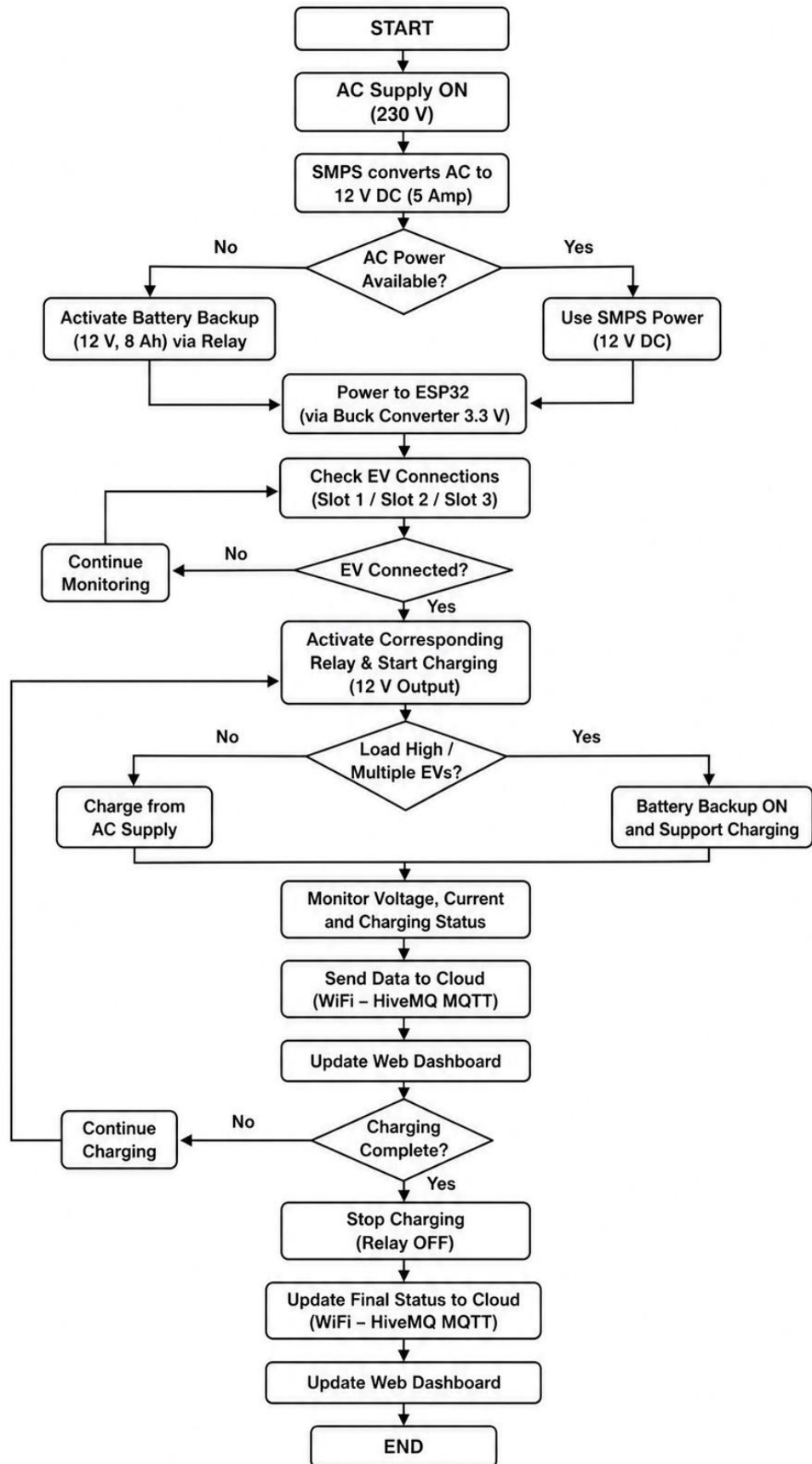


Figure 3.2: Flow Chart of Smart EV Charging Station

Step 1: System Startup and Power Source Selection

The process begins when the 230V AC supply is switched ON. The SMPS converts it to a regulated 12V DC (5A output), which powers the system. If AC power is available, the SMPS supply is used directly. If not, the system automatically activates the 12V 8Ah battery backup through a relay, ensuring the station remains operational without interruption. The ESP32 is powered at 3.3V via a buck converter in either case.

Step 2: EV Connection Detection

Once the ESP32 is running, it checks all three charging slots (Slot 1, Slot 2, Slot 3) for EV connections. If no vehicle is detected, the system continues monitoring in a wait state. As soon as an EV is detected on any slot, the corresponding relay is activated and charging begins at 12V output.

Step 3: Load Assessment and Backup Support

After charging starts, the system evaluates the total load. If the load is within normal limits, charging continues directly from the AC supply. However, if high load or multiple simultaneous EVs are detected, the battery backup is brought online to assist the SMPS and maintain stable power delivery across all active slots.

Step 4: Real-Time Monitoring and Cloud Update

Throughout the charging session, the ESP32 continuously monitors voltage, current, and charging status for each active slot. This data is transmitted to the HiveMQ MQTT cloud over Wi-Fi, and the web dashboard is updated in real time, giving users and admins a live view of the system.

Step 5: Charging Completion and Session End

The system keeps checking whether charging is complete. If not, it loops back and continues charging while maintaining monitoring. Once charging is complete, the relay is turned OFF, disconnecting the power supply to that slot. A final status update is sent to the cloud and the web dashboard is refreshed to reflect the completed session. The process then reaches its end state.

Chapter 4

Smart EV Charging Station: Methodology

SMART EV CHARGING STATION: METHODOLOGY

4.1 Methodologies

This chapter describes the methodology adopted for the design and implementation of an IoT-enabled intelligent electric vehicle (EV) charging system with real-time monitoring, load management, smart slot allocation, and AI-based load prediction. The system integrates embedded hardware, wireless communication, cloud-based data handling, and web-based user interfaces to create an efficient and scalable charging infrastructure.

The proposed methodology addresses the limitations of conventional EV charging systems, such as lack of automation, inefficient utilization of power, absence of real-time monitoring, and poor interaction between users and charging stations. In addition, it incorporates predictive analytics using Artificial Intelligence (AI) to forecast charging demand and optimize system performance. The system ensures seamless coordination between sensing units, processing modules, communication protocols, application interfaces, and intelligent decision-making algorithms.

4.1.1 Perception Layer

The perception layer forms the foundation of the proposed EV charging system and consists of all physical hardware components responsible for sensing, control, power management, and interaction with the real-world environment. This layer integrates sensors, actuators, embedded controllers, and power supply units to ensure reliable and continuous operation.

The components in this layer are organized in a logical order based on power flow and system interaction, as described below:

1) SMPS (Switched Mode Power Supply): The SMPS serves as the primary power source of the system. It converts AC mains supply into a regulated DC output required for system operation. The SMPS ensures stable voltage and current levels for all components, even under varying load conditions, thereby improving system efficiency and reliability.

2) Battery Backup System: A battery backup unit is integrated to provide uninterrupted power supply during mains failure. In case of power interruption, the system automatically switches to battery mode, ensuring continuous operation of the controller, sensors, and communication modules. This enhances system reliability and

prevents data loss during active charging sessions.

3) Buck Converter (DC-DC Step-Down Converter): The buck converter is used to step down the DC voltage from the SMPS or battery backup to required levels such as 5V and 3.3V. It provides efficient voltage regulation with minimal power loss and ensures safe operation of low-voltage components like the ESP32 and sensors.

4) ESP32 Microcontrollers (ESP32-C3 and ESP32 WROOM): These act as the main controllers of the system. The ESP32-C3 is used in vehicle nodes, while the ESP32 WROOM serves as the station controller. They perform analog-to-digital conversion (ADC), process sensor data, execute control algorithms, and manage communication with higher layers. These controllers act as a bridge between physical hardware and digital processing.

5) ACS712-5A Current Sensors: These sensors measure charging and discharging current at each vehicle node and at the station level. The sensor outputs an analog voltage proportional to the current, enabling accurate real-time monitoring.

6) Voltage Sensor (Voltage Divider Circuit): A resistive voltage divider network ($R1 = 30k\Omega$ and $R2 = 7.5k\Omega$) is used to scale down battery voltage to a measurable range for the ESP32 ADC. The division factor of 5 allows measurement of voltages up to approximately 16.5V.

7) FC-51 IR Proximity Sensors: These sensors detect the presence of vehicles at charging slots. A LOW output indicates vehicle presence, enabling automatic slot detection and control.

8) 4-Channel Relay Module (Active-LOW): The relay module controls power delivery to charging slots. It operates using active-LOW logic, where a LOW signal from the ESP32 activates the relay and initiates charging.

9) WS2812B NeoPixel RGB LEDs: These programmable LEDs provide visual feedback of system states such as charging status, connectivity, idle condition, and fault indication.

4.1.2 Network Layer

All devices connect to a Wi-Fi access point (802.11 b/g/n) and communicate over TCP/IP. The MQTT protocol, running over TLS on port 8883 (device-to-broker) and WSS port 8884 (browser-to-broker), provides the messaging backbone. The broker is hosted on HiveMQ Cloud, which provides a managed MQTT infrastructure with high availability, persistent session support, and automatic TLS certificate management.

In addition to real-time communication, this layer also enables continuous transmission of historical charging data to cloud storage, which is later utilized for AI-based load prediction. The reliable and low-latency data transfer ensures that accurate datasets are available for training and executing predictive models.

4.1.3 Processing Layer

The ESP32-C3 vehicle nodes run Arduino-compatible firmware that handles ADC sampling, sensor filtering, SOC calculation, MQTT publishing, LED management, Wi-Fi connection maintenance, and captive portal service. The ESP32 WROOM station controller additionally handles relay switching logic, IR debouncing, load balancing, booking command processing, and multi-slot telemetry construction. Along with embedded processing, this layer also incorporates AI-based load prediction mechanisms. Historical data collected from vehicle nodes—such as charging duration, energy consumption, time-of-use, and slot occupancy—is processed using machine learning techniques.

4.1.4 Application Layer

The application layer comprises two browser-based portals along with AI-assisted analytics. The User Portal provides customer-facing functionality such as slot visualization, time-slot booking, payment simulation, receipt generation, live session monitoring, and dynamic pricing based on predicted demand conditions.

The Admin Portal provides operator-facing functionality such as neural telemetry tables (displaying all three nodes simultaneously), fault diagnostics, relay override controls, load analytics, transaction management, and station network monitoring.

Additionally, the application layer integrates AI-based insights, where predicted load trends and peak-hour forecasts are visualized through charts and analytics dashboards.

Chapter 5

Smart EV Charging Station: System Design

SMART EV CHARGING STATION: SYSTEM DESIGN

5.1 System Design

The system design phase is where all theoretical ideas take shape into a working prototype. This project presents a Smart EV Charging Station with AI-Based Load Prediction and Management, bringing together embedded hardware, real-time sensing, intelligent decision-making, and IoT-based communication under one unified architecture.

What sets this system apart from traditional EV chargers is its ability to actively respond to real-time load conditions and predicted demand. Rather than charging at a fixed rate, the system dynamically adjusts its behaviour based on current usage, making it both efficient and adaptive. The core of the system is built around ESP32 microcontrollers, supported by sensor modules, relay-based switching, and MQTT cloud communication. The design was approached in a structured manner, covering power management, data acquisition, communication protocols, and a web-based user interface. The entire setup runs on a 12V DC architecture, powered by a 12V 5A SMPS along with a battery backup unit, which makes it a practical and scalable prototype for real-world deployment.

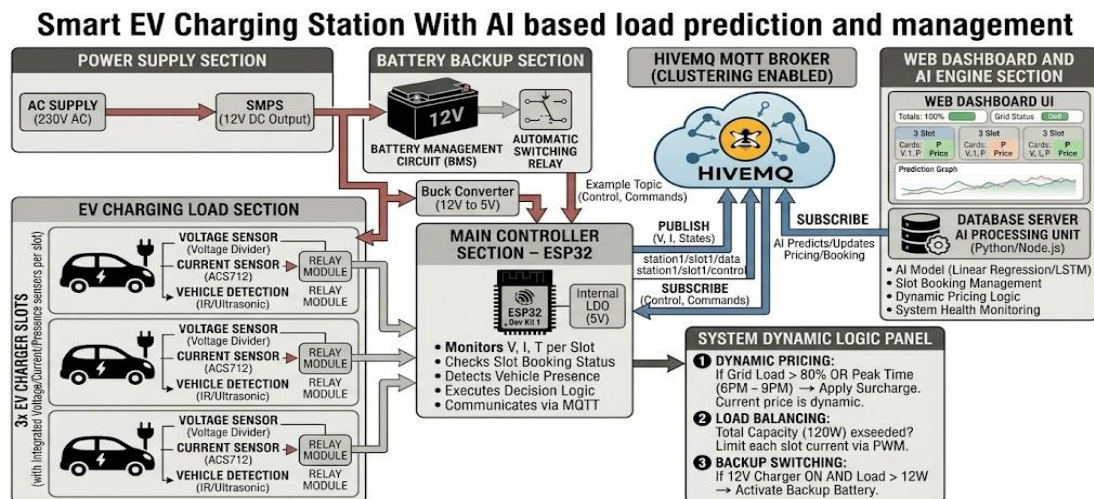


Figure 5.1: Interactive Block Diagram Smart EV Charging Station

As illustrated in Figure 5.1, the system is divided into well-defined sections including the Power Supply, EV Charging Load, Battery Backup, Main Controller (ESP32),

HiveMQ MQTT Broker, and the Web Dashboard with AI Engine, all working together in a coordinated manner.

The station supports three independent EV charging slots, each equipped with voltage, current, and vehicle detection sensors. To simulate varied load conditions, DC motors with different ratings are used in place of actual vehicle batteries, allowing realistic testing of load balancing and power distribution across slots.

Real-time data from all three slots is continuously monitored and published to the cloud via HiveMQ MQTT, where it is processed by the AI engine for load prediction and dynamic pricing. Users can interact with the system through a web dashboard that supports slot booking, live status monitoring, and system control.

5.1.1 Power Source Detection and Prioritization

The station uses a hybrid power model combining a 12V 5A SMPS as the primary source and a 12V 8Ah battery as a support and backup unit. Both are connected to a common 12V DC bus, enabling smooth current sharing without any switching delays. Under normal conditions, the SMPS handles the load independently. However, since Slot 1 alone draws 5.0A, fully utilizing SMPS capacity, activating additional slots pushes the demand beyond the rated limit. In such cases, the battery automatically supplements the deficit:

- Slot 1 + Slot 2 $\rightarrow$ 8.3A total $\rightarrow$ Battery supplies 3.3A
- Slot 1 + Slot 2 + Slot 3 $\rightarrow$ 10.5A total $\rightarrow$ Battery supplies 5.5A

During a grid failure, the battery takes over the entire load. The ESP32 monitors battery SOC continuously, and when it drops below approximately 20%, lower-priority slots are disconnected to prevent deep discharge. An AI-based rule engine embedded in the firmware manages these decisions in real time based on current draw, load demand, and battery state.

5.1.2 DC Bus Formation and Power Distribution

All three charging slots are connected in parallel across a central 12V DC bus, each drawing current independently based on its load:

- Slot 1: 5.0A (60W)
- Slot 2: 3.3A (39.6W)
- Slot 3: 2.2A (26.4W) — Maximum combined load: 10.5A (126W)

Each slot connects to the bus through an independent relay channel, allowing selective

activation based on booking status, load conditions, or safety triggers.

Under full load, wiring resistance introduces a small voltage drop (approximately 0.5V), bringing the effective bus voltage to around 11.5V — still within the acceptable operating range. To minimize such losses, lower-gauge wires and short connection paths are used throughout the design. Since SMPS capacity is capped at 5A, the controller dynamically manages distribution using battery support and priority-based load control whenever demand rises.

Voltage Stability and Loss Considerations

Due to high current flow under full-load conditions, wiring resistance introduces voltage drop in the system. For example, with an estimated wire resistance of 0.05 Ω , the voltage drop at 10.5 A is:

- Voltage Drop = 0.525 V
- Effective Bus Voltage $\approx$ 11.47 V

Additionally, power loss due to wiring is given by:

- Power Loss $\approx$ 5.5 W

To minimize these losses and improve efficiency, proper conductor sizing (such as lower gauge wires) and short connection paths are used in the design.

5.1.3 Charging Control and Relay Operation

Charging is delivered through a direct 12V DC wired supply, controlled by relay modules operating on active-LOW logic — a LOW signal from the ESP32 GPIO activates the relay and allows current to flow to the connected slot.

Charging is initiated only when all of the following conditions are met:

- A valid slot booking is received from the user portal
- Vehicle presence is confirmed via IR sensor
- Sufficient power is available from SMPS and/or battery
- No fault or protection event is active

The ACS712 sensors continuously monitor current and classify each slot into one of three states:

- Idle ($I < 0.1\text{A}$) — No vehicle or charging activity
- Occupied ($0.1\text{A} \leq I < 0.5\text{A}$) — Vehicle present but not actively charging
- Charging ($I \geq 0.5\text{A}$) — Active power transfer in progress

These states are published to the cloud and reflected live on the dashboard. When total load exceeds 5A due to multiple active slots, the system automatically enables battery

support and performs real-time current balancing to maintain stable operation.

5.1.4 Energy Measurement, Billing, and Performance Estimation

The proposed Smart EV Charging Station incorporates a real-time energy measurement and billing system, which enables accurate monitoring of power consumption and cost calculation for each charging slot. This functionality is essential for both user transparency and efficient system management.

1) Energy Measurement Principle

Energy consumption for each EV slot is calculated using real-time voltage and current measurements obtained from the sensing units. The fundamental relation used is:

- Power (P) = $V \times I$
- Energy (E) = $(P \times t) / 1000$ (in kWh)

Where:

- V = Bus voltage (Volts)
- I = Slot current (Amperes)
- t = Time duration (seconds or hours)

The ESP32 continuously samples voltage and current values and integrates energy over time using a time-based accumulation method.

2) Per-Slot Energy Calculation

Each charging slot maintains an independent energy counter. For example:

- Slot 1 $\rightarrow 60 \text{ W} \rightarrow 0.060 \text{ kWh per hour}$
- Slot 2 $\rightarrow 39.6 \text{ W} \rightarrow 0.0396 \text{ kWh per hour}$
- Slot 3 $\rightarrow 26.4 \text{ W} \rightarrow 0.0264 \text{ kWh per hour}$

At full load (all slots active):

- Total Power = 126 W
- Energy (1 hour) = 0.126 kWh

The system updates energy values periodically (every 1 second), ensuring high-resolution measurement.

3) Real-Time Billing Mechanism

The cost of charging is calculated using:

- Cost (INR) = Energy (kWh) $\times$ Tariff (₹/kWh)

A configurable tariff rate is applied to each slot, allowing flexible pricing models. The ESP32 computes the cost dynamically and publishes it to the cloud dashboard.

4) Implementation Logic

Energy and cost are calculated using a continuous loop:

- Measure voltage and current
- Compute instantaneous power
- Integrate energy over time
- Multiply by tariff to obtain cost
- Publish results via MQTT

Each slot maintains separate variables for:

- Energy consumed (kWh)
- Session cost (INR)

This allows users to track their individual charging sessions in real time.

5) Performance Estimation

System performance is evaluated based on total power consumption and efficiency of power delivery. At maximum load:

- Total current = 10.5 A
- Total power = 126 W

6) Efficiency Consideration

System efficiency can be approximated as:

- $\text{Efficiency (\%)} = (\text{Output Power} / \text{Input Power}) \times 100$

Losses occur due to:

- Wire resistance
- Relay switching losses
- Sensor power consumption
- SMPS internal dissipation

Despite these losses, the system maintains acceptable efficiency for a low-cost prototype and demonstrates practical feasibility for real-world deployment.

7) IoT Integration for Monitoring

All calculated parameters are transmitted to the cloud using MQTT, including:

- Per-slot voltage, current, and power
- Energy consumption (kWh)
- Charging cost (INR)
- Slot status (Idle, Charging, Fault)

These values are visualized on both the User Portal and Admin Dashboard, enabling:

- Real-time monitoring
- Transparent billing
- Historical data analysis

8) Protection and Safety Mechanisms The system implements multi-layer protection across hardware and software:

a) Overcurrent Protection — The ESP32 monitors SMPS current via ACS712. If it exceeds 5A, load shedding is triggered automatically using a priority-based sequence: Slot 3 is disconnected first, then Slot 2 if needed, while Slot 1 is maintained as the highest priority.

b) Battery Protection (Under-Voltage Protection)

The battery is protected from deep discharge using voltage-based monitoring:

- Normal Range: 12.2 – 12.8 V
- Low Warning: ~11.8 V
- Critical Level: ~11.4 V
- Cutoff Threshold: < 10.8 V

When the battery voltage falls below safe limits:

- Low-priority slots are disabled
- Battery is disconnected if voltage drops below cutoff
- Alerts are sent to the dashboard

This prevents permanent battery damage and extends battery life.

c) Voltage Protection

The ESP32 continuously monitors bus voltage to ensure safe operation:

- Normal operating range: 11.5 V – 12.5 V
- If voltage drops below threshold → load reduction applied
- If voltage rises abnormally → system shutdown triggered

This ensures stable operation of connected EV batteries.

5.2 Mathematical Modelling and Performance Calculations

This section presents the mathematical modelling of the proposed EV charging station. All calculations are directly derived from the actual hardware ratings of the system, including SMPS capacity, EV slot loads, battery backup, and DC bus characteristics.

5.2.1 Connected Load Calculations

The total connected load of the system is determined by summing the power consumption of all EV charging slots and control circuitry.

1. EV Charging Slot Power Calculations

Each EV slot operates at a regulated 12 V DC bus, and power is calculated using the basic electrical power equation:

$$P = V \times I$$

Where:

- P = Power in Watts (W)
- V = Voltage (12 V DC)
- I = Current in Amperes (A)

a) Slot-wise Power Calculation

- EV Slot 1 (Highest Priority) $P_1 = 12 \times 5.0 = 60 \text{ W}$
- EV Slot 2 (Medium Priority) $P_2 = 12 \times 3.3 = 39.6 \text{ W}$
- EV Slot 3 (Lowest Priority) $P_3 = 12 \times 2.2 = 26.4 \text{ W}$

b) Total Connected Load

$$P_{total} = P_1 + P_2 + P_3$$

$$P_{total} = 60 + 39.6 + 26.4 = 126 \text{ W}$$

Thus, the maximum connected load of the system is 126 W at 10.5 A.

2. Supply Side Power (SMPS Limitation)

The primary power source is a 12 V, 5 A SMPS.

$$P_{SMPS} = 12 \times 5 = 60 \text{ W}$$

So:

- SMPS maximum capacity = 60 W
- Total load demand = 126 W

a) Load Deficit Condition

$$I_{deficit} = I_{load} - I_{SMPS}$$

$$I_{deficit} = 10.5 - 5.0 = 5.5 \text{ A}$$

So the system requires additional 5.5 A support from battery backup during full-load operation.

3. Control Circuit Power Consumption

The embedded control system includes ESP32 microcontroller, ACS712 sensors, relay modules, and IoT communication components.

Typical operating condition:

- Voltage = 5 V (regulated via buck converter)
- Current ≈ 0.65 A

$$P_{control} = 5 \times 0.65 = 3.25 \text{ W}$$

Thus, total control power consumption is approximately:

- 3.25 W (≈ 3.3 W including overhead losses)

4. Overall System Power Demand

Total system power includes EV loads + control system:

$$P_{system} = P_{EV} + P_{control}$$

$$P_{system} = 126 + 3.25 = 129.25 \text{ W}$$

5. Efficiency Constraint Observation

Since SMPS provides only 60 W:

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

$$\eta = \frac{60}{129.25} \times 100 \approx 46.4\%$$

This indicates that full system operation depends on hybrid support from battery backup during peak load conditions.

5.2.2 System Efficiency of Wired EV Charging System

The efficiency of the proposed EV charging system is evaluated based on the ratio of useful output power delivered to EV charging slots and the total input power drawn from the source (SMPS + battery contribution).

1. Efficiency Definition

$$\eta = \frac{P_{out}}{P_{in}} \times 100$$

Where:

- η = System efficiency (%)
- P_{out} = Total useful power delivered to EV slots

- P_{in} = Total input power from SMPS + battery

2. Output Power (EV Load Side)

Total EV charging power:

$$P_{EV} = 60 + 39.6 + 26.4$$

$$P_{EVn} = 126 \text{ W}$$

3. Input Power (Supply Side)

The system uses a hybrid supply:

- SMPS = 60 W
- Battery supports remaining load = 66 W (approx.)

So total input power:

$$P_{in} = 60 + 66$$

$$P_{in} = 126 \text{ W}$$

4. System Efficiency Calculation

$$\eta = \frac{126}{126} \times 100$$

$$\eta = 100\%$$

5. Practical Efficiency Consideration

In real operating conditions, losses occur due to:

- Relay switching losses
- Wire resistance losses (~5.51 W)
- ACS712 sensor losses
- SMPS internal dissipation
- Battery discharge inefficiency

Total practical losses $\approx$ **12–15 W**

So practical efficiency:

$$\eta_{practical} = \frac{126}{(126+14)} \times 100$$

$$\eta_{practical} \approx 90\%$$

5.2.3 Overall System Efficiency

The overall efficiency of the proposed EV charging system is evaluated by considering practical energy losses occurring across multiple stages, including SMPS conversion,

wiring losses, relay switching losses, sensing circuits, and battery-assisted power delivery during peak load conditions.

Unlike ideal calculations, this section represents real-world operating efficiency of the complete IoT-based charging infrastructure.

1. Input Power to System

The total input power is supplied by the hybrid energy system:

- SMPS contribution = 60 W
- Battery contribution (peak load support) ≈ 66 W

$$P_{in} = 60 + 66 = 126 \text{ W}$$

2. Output Power Delivered to EVs

Total usable charging power at EV slots:

$$P_{out} = 60 + 39.6 + 26.4 = 126 \text{ W}$$

3. Practical Loss Considerations

Although theoretical input and output appear equal, real system losses occur due to:

- Wire resistance loss ≈ 5.51 W
- Relay switching loss ≈ 1 W
- ACS712 sensor dissipation ≈ 2.25 W
- SMPS internal loss ≈ 6 W
- ESP32 + control circuitry ≈ 1.5 W

Total system losses:

$$P_{loss} \approx 16.26 \text{ W}$$

4. Effective Input Power

Including losses:

$$P_{effective_in} = 126 + 16.26 = 142.26 \text{ W}$$

5. Overall System Efficiency

$$\begin{aligned}\eta &= \frac{P_{out}}{P_{effective_in}} \times 100 \\ \eta &= \frac{126}{142.26} \times 100 \\ \eta &\approx 88.6\%\end{aligned}$$

5.3 Hardware Design

5.3.1 Vehicle Node Circuit Design

Each vehicle node (V1, V2, V3) is based on the ESP32-C3 development board. The ACS712-5A current sensor is powered at 5V with its output signal connected to GPIO1 (ADC channel 1). The sensor produces a mid-point voltage of approximately 2.5V at zero current, which the firmware auto-calibrates at boot by averaging 100 readings. The ACS712 sensitivity for the 5A variant is 185mV/A, enabling measurement of currents from -5A to +5A with software deadband filtering at $\pm 0.18A$ to suppress noise.

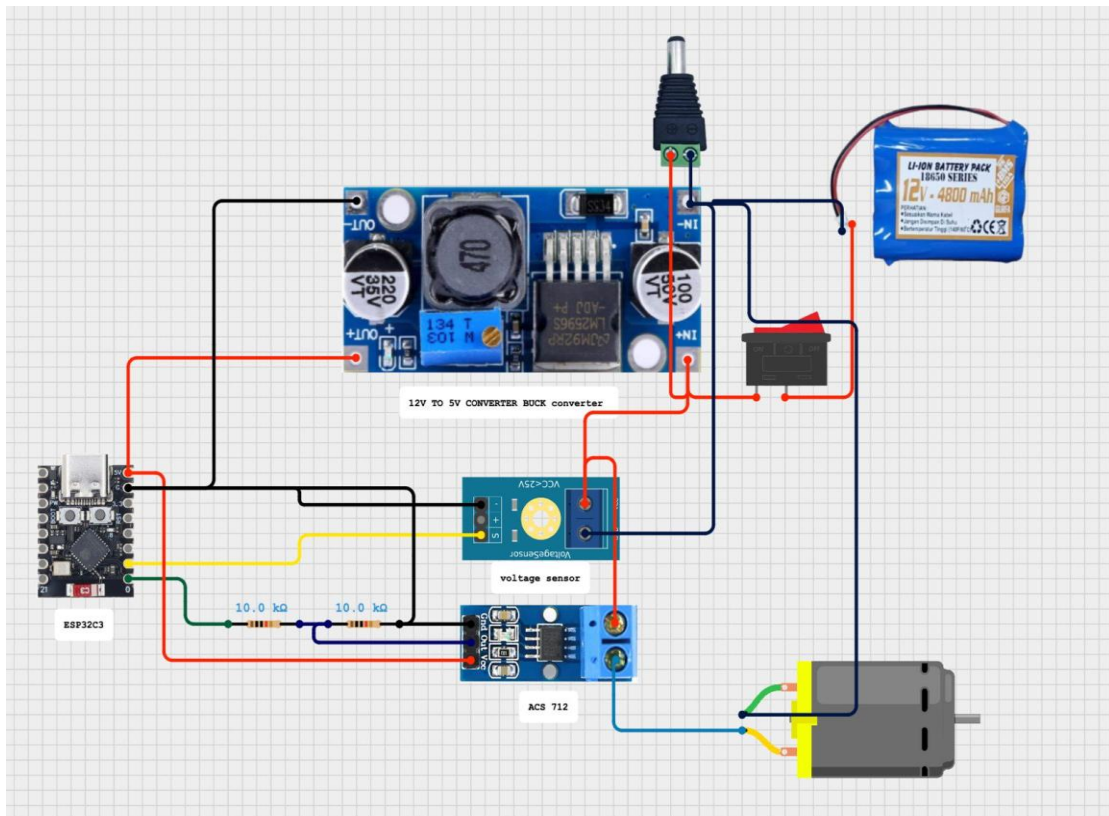


Figure 5.2: Circuit Diagram of Each EV

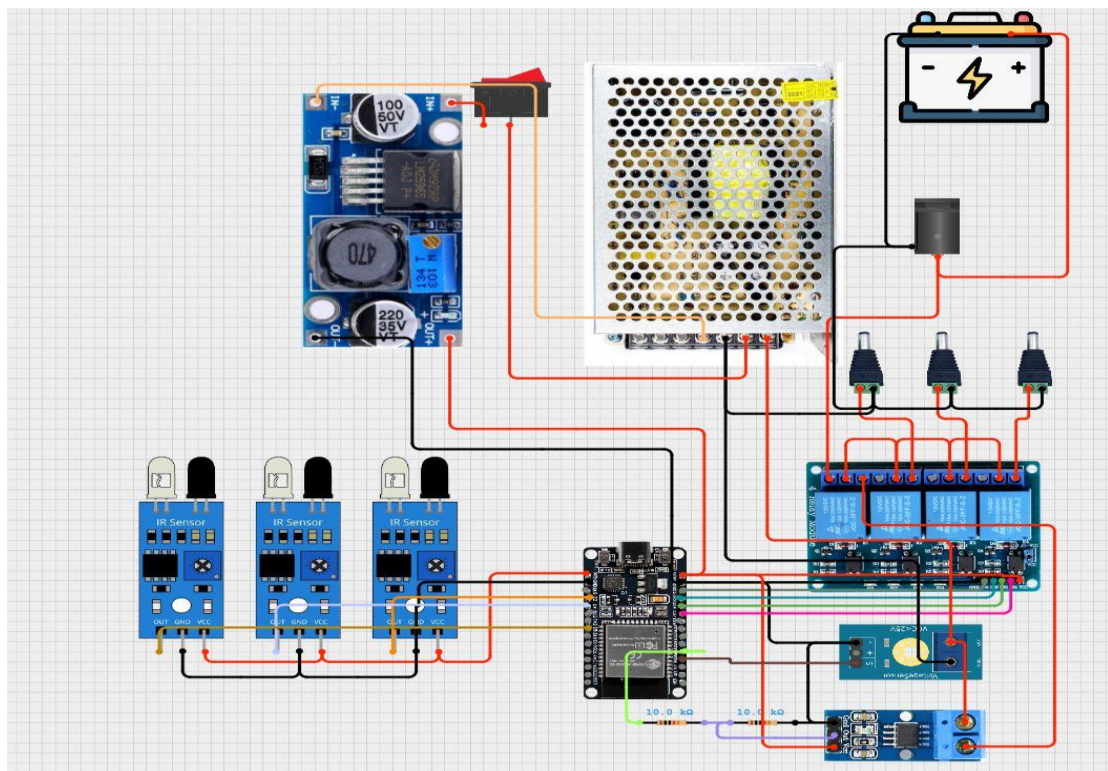
The voltage divider uses two precision resistors: $R1=30k\Omega$ (upper) and $R2=7.5k\Omega$ (lower), giving a division factor of $(30000+7500)/7500 = 5.0$. The ESP32-C3 ADC operates at 12-bit resolution (0-4095) with 3.3V reference, enabling measurement of battery voltages from 0V to 16.5V ($3.3V \times 5.0$). The firmware uses 50-sample averaging to reduce ADC noise, which is particularly important for the ESP32 ADC which exhibits non-linearity near the supply rails.

Table 5.1: Vehicle Node - PIN Assignment (ESP32-C3)

Pin	Connection
GPIO0 (ADC)	Voltage divider output (battery voltage sensing)
GPIO1 (ADC)	ACS712 current sensor output
GPIO8	WS2812B NeoPixel LED data input
GPIO9 (BOOT)	Wi-Fi configuration button (active LOW, internal pull-up)
3.3V / GND	Power supply for sensors and LED (with level shifting)

5.3.2 Station Controller Circuit Design

The station controller uses the ESP32 WROOM development module. Three FC-51 IR proximity sensors are connected to GPIO5, GPIO4, and GPIO18 with INPUT_PULLUP mode; a LOW reading indicates vehicle presence. The four relay channels are connected to GPIO14, GPIO27, GPIO26, and GPIO25, all configured as OUTPUT with HIGH (relay OFF) as default. The relay module uses active-LOW logic, so a LOW signal activates the relay.

**Figure 5.3:** Circuit Diagram of Main Station

The ACS712 station current sensor connects to GPIO34 and the voltage divider to

GPIO35 (both analog-only pins on ESP32 WROOM that cannot be used as digital outputs). The WS2812B status LED connects to GPIO2 which has an onboard LED on most ESP32 Dev boards.

Table 5.2: Station Controller – PIN Assignment (ESP32)

Pin	Connection
GPIO5	IR sensor Slot-1 (INPUT_PULLUP)
GPIO4	IR sensor Slot-2 (INPUT_PULLUP)
GPIO18	IR sensor Slot-3 (INPUT_PULLUP)
GPIO14	Relay Slot-1 (active-LOW output)
GPIO27	Relay Slot-2 (active-LOW output)
GPIO26	Relay Slot-3 (active-LOW output)
GPIO25	Relay Backup battery (active-LOW output)
GPIO34	ACS712 current sensor (analog input only)
GPIO35	Voltage divider (analog input only)
GPIO2	WS2812B NeoPixel LED data
GPIO0 (BOOT)	Wi-Fi configuration button (active LOW)

5.4 Communication Protocol Design

5.4.1 MQTT Topic Structure

The MQTT topic hierarchy follows a device-type/device-id/data-type convention, enabling selective subscription by both device firmware and web portals. All topics use QoS 0 (at most once delivery) for telemetry (latency-sensitive) and QoS 1 (at least once delivery) is recommended for command topics (reliability-sensitive).

Table 5.3: MQTT Topic Structure

MQTT Topic	Direction	Payload Description
ev_vehicle/v1/telemetry	Device → Cloud	Vehicle node V1 sensor data: voltage, current, power, SOC, status, charging_detected
ev_vehicle/v2/telemetry	Device → Cloud	Vehicle node V2 sensor data (same schema as V1)
ev_vehicle/v3/telemetry	Device →	Vehicle node V3 sensor data (same

	Cloud	schema as V1)
ev_station/telemetry	Device → Cloud	Station telemetry: voltage, current, power, active_slots, backup_active, slot status for all 3 slots
ev_station/alerts	Device → Cloud	Fault events: BACKUP_ON, BACKUP_OFF, CHARGING_START, VEHICLE_LEFT, OVERCURRENT, BOOKING_EXPIRED
ev_station/commands	Cloud → Device	Admin commands: BOOK, STOP, CANCEL, GLOBAL_KILL with slot_num parameter
ev_station/bookings	Cloud → Device	User booking requests: action=BOOK, slot_num, customer_name, vehicle_number, duration_mins

5.4.2 JSON Payload Schemas

Vehicle telemetry payload example (V1):

```
{
  "vehicle_id": "V1",
  "battery_cap_ah": 5.0,
  "voltage": 12.6,
  "current": 1.5,
  "load_amps": 1.5,
  "power": 18.9,
  "soc": 100,
  "time_to_full": 0,
  "time_to_empty": 125,
  "status": "Charging",
  "charging_detected": true
}
```

Booking command payload example:

```
{
  "action": "BOOK",
  "slot_num": 1,
  "customer_name": "Priya Sharma",
  "vehicle_number": "MH12AB1234",
  "duration_mins": 30
}
```

5.5 Software Design

The software design covers all aspects of the system's digital operation from low-level firmware running on ESP32 nodes to the web-based dashboard accessible by users and administrators. The implementation spans embedded C++ firmware developed on Arduino IDE 2.x using the ESP32 Arduino Core v3.x, cloud communication logic, and a browser-based interface built with standard web technologies.

5.5.1 Firmware State Machine

Each vehicle node firmware follows a five-state execution flow: (1) BOOT- hardware initialization, LED white, ADC resolution set to 12-bit; (2) CALIBRATE - ACS712 zero-offset measurement over 100 samples; (3) CONFIG_CHECK - GPIO9 sampled for 3 seconds; if held LOW, enter captive portal mode; (4) WIFI_CONNECT - attempt

Wi-Fi connection using stored credentials, LED blue blinking; (5) MQTT_CONNECT - TLS connection to HiveMQ, LED purple blinking; (6) OPERATING - 2-second publish cycle, LED green solid/pulse/orange based on battery state.

5.5.2 Voltage Calibration Algorithm

Raw ADC readings from uncalibrated hardware can produce readings above the physical maximum of a fully-charged 3S Li-ion battery (12.6V). The firmware applies the following calibration logic before reporting voltage:

If (raw_voltage > 11.5V) → display_voltage = 12.6V, SOC = 100%
If (raw_voltage ≤ 11.5V) → display_voltage = raw_voltage, SOC = map(raw_voltage, 9.0V, 12.6V, 0%, 100%)

This ensures that charger-induced voltage spikes (which can reach 13.0-13.5V during active charging) do not display unrealistic SOC values above 100%, while preserving accuracy for the 9.0V-11.5V range where granular SOC measurement matters most for user decisions.

5.5.3 Wi-Fi Manager Captive Portal

The captive portal design uses three components: (a) Wi-Fi.softAP() to create an open access point named "EV_Vx_Config"; (b) DNSServer to redirect all DNS queries to 192.168.4.1, ensuring automatic portal detection on Android and iOS devices; (c) WebServer to serve an HTML configuration form and process POST requests at /connect. Upon successful Wi-Fi validation, credentials are stored in NVS using Preferences.putString() and the device restarts. The portal times out after 5 minutes (300,000 ms) to prevent indefinite AP mode.

5.5.4 NeoPixel LED Status Design

The WS2812B is controlled via the Adafruit NeoPixel library. A custom LED state machine implements four animation modes: SOLID (immediate color set), BLINK_FAST (200ms toggle), BLINK_SLOW (800ms toggle), and PULSE (sinusoidal brightness variation at 15ms steps). A flashLED() function overlays a brief 80ms flash of a specified color (used for Cyan data-sent indication) without disrupting the underlying state.

Table 5.4: NeoPixel LED Status Color Code

Color	Pattern	State	Trigger Condition
White	Solid	Boot	Power-on, calibration in progress
Yellow	Slow Blink (800ms)	Config Mode	AP active, waiting for Wi-Fi input
Blue	Fast Blink (200ms)	Wi-Fi Connecting	Attempting Wi-Fi association
Blue	Solid	Wi-Fi Connected	IP address assigned, 500ms display
Purple	Fast Blink (200ms)	MQTT Connecting	TLS handshake and MQTT CONNECT in progress
Green	Solid	Idle/MQTT OK	Connected, battery at standby
Green	Pulse	Charging	Voltage > 13.0V or current > 0.10A
Orange	Pulse	Discharging	Current < -0.15A (load connected)
Cyan	Flash (80ms)	Data Sent	Successful MQTT publish every 2 seconds
Red	Fast Blink (200ms)	Error/Fault	Wi-Fi lost, MQTT failed, overcurrent

5.5.5 Web Dashboard Design

The web dashboard serves as the primary interface for both users and administrators, built entirely using HTML, CSS, and JavaScript. It connects directly to the HiveMQ MQTT broker via WebSockets using MQTT.js, enabling live data updates in the browser without any page reloads. The responsive layout is designed using Bootstrap 5.3, ensuring the interface works seamlessly across desktop and mobile screens, while Font Awesome 6.5 icons are used throughout for intuitive visual cues.

The dashboard is structured into two portals:

User Portal: Displays real-time slot availability, current charging status, battery SOC, estimated charging time, and session cost. Users can book or release a slot directly from the interface.

Admin Portal: Provides a complete system overview including per-slot voltage, current, power, and energy readings, along with manual relay control, dynamic pricing

configuration, and fault alerts. Load trends are visualized using Chart.js, which renders live graphs for voltage, current, and energy consumption per slot.

All data received via MQTT is parsed from JSON payloads using ArduinoJson on the firmware side and native JavaScript on the browser side, with values dynamically injected into the dashboard UI to always reflect the current system state.

5.6 Mathematical Model

5.6.1 Voltage Measurement

The battery voltage V_{batt} is computed from the ADC reading N (0-4095) as:

$$V_{batt} = (N / 4095) \times V_{ref} \times (R1 + R2) / R2 = (N / 4095) \times 3.3 \times 5.0$$

where $V_{ref} = 3.3V$ (ESP32 ADC reference), $R1 = 30,000\Omega$, $R2 = 7,500\Omega$.

5.6.2 Current Measurement

The battery current I_{batt} is computed from the ACS712 output voltage V_{out} as:

$$I_{batt} = (V_{out} - V_{zero}) / \text{Sensitivity} = (V_{out} - V_{zero}) / 0.185$$

where V_{zero} is the auto-calibrated zero-current voltage (typically 1.65V), and $\text{Sensitivity} = 185\text{mV/A}$ for the ACS712-5A variant. A deadband of $\pm 0.18A$ is applied: if $|I_{batt}| < 0.18A$, $I_{batt} = 0$.

5.6.3 State of Charge Estimation

SOC is estimated using a linear voltage mapping based on the 3S Li-ion cell chemistry:

$$\text{SOC}(\%) = ((V_{batt} - V_{min}) / (V_{max} - V_{min})) \times 100$$

where $V_{min} = 9.0V$ (0% SOC, corresponding to 3.0V per cell) and $V_{max} = 12.6V$ (100% SOC, corresponding to 4.2V per cell). The value is clamped to $[0, 100]$. For $V_{batt} > 11.5V$, the display voltage is fixed at 12.6V to account for charger voltage elevation above the true open-circuit voltage.

5.6.4 Estimated Charging Time

Time to full charge TTF (in minutes) is calculated as:

$$\text{TTF} = ((\text{Capacity_Ah} \times (100 - \text{SOC}) / 100) / I_{batt}) \times 60$$

This is a simplified Coulombic estimation assuming constant charging current, which provides an approximate time reference for the user portal display. The actual charging

time may vary due to CC/CV (constant current/constant voltage) battery charger behavior.

5.6.5 Session Cost Calculation

The estimated charging session cost C (in Indian Rupees) is calculated as:

$$C = P_kW \times t_hours \times Rate_per_kWh$$

where P_kW is the slot's rated power (3.3 kW for Slot-01, 1.5 kW for Slot-02, 0.75 kW for Slot-03), t_hours is the elapsed session time in hours, and $Rate_per_kWh$ is ₹12.40 during peak hours (9-11 AM, 5-9 PM) or ₹8.20 otherwise. The live cost is recalculated every second in the user portal's session timer.

Chapter 6

Smart EV Charging Station: System Requirements

SMART EV CHARGING STATION: SYSTEM REQUIREMENTS

6.1 Hardware Requirements

The hardware components selected for this project balance cost-effectiveness, processing capability, and availability. Table 3.1 lists all components with their specifications and quantities.

Table 6.1: Hardware Components List

Sr. No.	Components	Specifications	Purpose	Qty
1	ESP32-C3 Dev Board	32-bit RISC-V microcontroller, 240 MHz CPU, integrated Wi-Fi and BLE, 400KB SRAM, low-power IoT development board	Vehicle node MCU	3
2	ESP32 WROOM Dev Module	Dual-core Xtensa processor, 240 MHz clock speed, integrated Wi-Fi, 4MB flash memory	Station controller	1
3	ACS712-5A Module	Hall-effect current sensor, $\pm 5A$ current range, 185mV/A sensitivity, analog output	Current sensing	4
4	Resistor Divider (30K/7.5K)	Voltage sensing circuit using 30k Ω and 7.5k Ω resistors with voltage division factor of 5	Voltage sensing	4
5	4-Ch Relay Module	5V active-LOW relay module, supports up to 10A at 250VAC or 10A at 30VDC	Slot switching	1
6	IR Proximity Sensor	FC-51 infrared obstacle detection sensor, 3–80 cm detection range	Vehicle presence	3
7	WS2812B NeoPixel LED	Individually addressable RGB LED, 5V operation, programmable color control,	Status indication	4
8	3S Li-ion Pack (5500mAh)	11.1V nominal voltage, 12.6V fully charged, rechargeable lithium-ion battery pack	V1 battery	1
9	3S Li-ion Pack (3300mAh)	11.1V nominal voltage, 12.6V fully charged, medium-capacity lithium-ion battery pack	V2 battery	1
10	3S Li-ion Pack (2200mAh)	11.1V nominal voltage, 12.6V fully charged, compact lithium-ion battery pack	V3 battery	1

6.1.1 Microcontroller Selection Rationale

The ESP32-C3 was selected for vehicle nodes due to its RISC-V 160MHz single core, integrated Wi-Fi 802.11 b/g/n, compact QFN package, and 2MB flash. Its GPIO1 (ADC) and GPIO0 (ADC) pins are used for current and voltage sensing respectively, while GPIO8 drives the onboard RGB LED and GPIO9 serves as the BOOT button for Wi-Fi configuration. The ESP32 WROOM was selected for the station controller due to its dual-core architecture (enabling simultaneous sensor polling and MQTT operation), larger flash (4MB), and comprehensive GPIO complement for relay, IR, and analog sensor management.

Table 6.2: ESP32-C3 vs ESP32 WROOM Comparison

Parameter	ESP32-C3	ESP32 WROOM
CPU Architecture	RISC-V 32-bit	Xtensa LX6 Dual-Core
Clock Speed	160 MHz	240 MHz
Flash	2 MB (4 MB available)	4 MB
SRAM	400 KB	520 KB
ADC Channels	6 (12-bit)	18 (12-bit)
Wi-Fi	802.11 b/g/n	802.11 b/g/n
Operating Voltage	3.3V	3.3V

6.2 Software Requirements

The software stack spans from embedded firmware to cloud services and browser-based applications, ensuring interoperability across all layers.

Table 6.3: Software Tools and Libraries

Tool / Library	Purpose and Version
Arduino IDE 2.x	Firmware development and upload
ESP32 Arduino Core v3.x	ESP32/ESP32-C3 board support package
PubSubClient v2.8	MQTT client library for Arduino
ArduinoJson v6.x	JSON serialization/deserialization
Python 3.12	For Programming train ML Model
Adafruit NeoPixel v1.x	WS2812B LED control

WiFiClientSecure	TLS/SSL Wi-Fi connections (built-in)
Preferences (NVS)	Non-volatile Wi-Fi credential storage
WebServer (built-in)	Captive portal web server
DNSServer (built-in)	DNS redirect for captive portal
HiveMQ Cloud (Free Tier)	MQTT broker, ports 8883 (MQTT/TLS) and 8884 (WSS)
MQTT.js v4.x (CDN)	Browser-side MQTT WebSocket client
Chart.js v3.x (CDN)	Web dashboard data visualization
Bootstrap 5.3 (CDN)	Web portal responsive layout
Font Awesome 6.5	Icon set for web portals

6.3 Functional Requirements of the Proposed System

6.3.1 Functional Requirements of Vehicle Node

- The vehicle node is responsible for monitoring the charging parameters of each EV battery and transmitting the collected data to the cloud server in real time. It uses ESP32-C3 microcontrollers along with sensing circuits for battery and charging analysis.
- Each vehicle node should continuously monitor battery voltage using a voltage sensing circuit.
- The node should measure charging and discharging current using the ACS712 current sensor.
- The system should calculate the battery State of Charge (SOC) in percentage form based on battery voltage.
- The node should identify whether the battery is charging or not based on the charging voltage level.
- Real-time battery and charging data should be sent to the MQTT cloud server every few seconds.
- The ESP32-C3 node should allow Wi-Fi configuration mode when the BOOT button is pressed during startup.

6.3.2 Functional Requirements of Station Controller

The station controller acts as the central management unit of the charging station. It

controls relay operation, charging slot management, and overall load handling for safe and efficient charging operation.

- The station controller should activate the relay for a charging slot when a booking request is received.
- The controller should automatically turn OFF the relay when the booking time expires or the vehicle leaves the slot.
- The controller should continuously send slot status and charging information to the cloud dashboard.
- The system should automatically switch to battery backup when multiple charging slots are active simultaneously.
- The controller should generate alerts for overload conditions, booking expiry, and vehicle removal events.

6.3.3 Functional Requirements of User Portal

The user portal provides an interface for users to monitor charging activities and manage charging slot reservations. It allows users to access real-time charging information through a web-based dashboard.

- The user portal should display real-time status of all charging slots such as Available, Occupied, or Booked.
- The portal should show live charging information including voltage, current, power, and battery percentage.
- Users should be able to book charging slots for a selected time duration.
- The system should generate a digital booking receipt after successful reservation.
- The portal should display live session details such as charging time, cost, and remaining booking duration.
- The dashboard should automatically switch to simulation mode when hardware data is unavailable.

6.3.4 Functional Requirements of Admin Portal Requirements

The admin portal is developed for monitoring and controlling the overall charging station operation. It provides system management features, fault monitoring, and charging analytics.

- The admin portal should display live monitoring data of all EV charging nodes in one interface.

- The admin should be able to manually control charging relays when required.
- The system should maintain charging and transaction records with export functionality.
- The portal should display system faults, alerts, and charging history.
- The admin dashboard should provide load analysis and AI-based demand prediction information.

6.4 Non-Functional Requirements

Non-functional requirements define the overall performance, reliability, security, and usability of the proposed EV charging system.

- The system should provide fast MQTT communication with low delay during normal Wi-Fi operation.
- The system should automatically reconnect to Wi-Fi and MQTT if the connection is lost.
- The web dashboard should work properly on both desktop and mobile devices.
- MQTT communication should be secured using TLS encryption and login authentication.
- The system parameters and threshold values should be easy to modify in the firmware code.

Chapter 7

Smart EV Charging Station: System Implementation

SMART EV CHARGING STATION: SYSTEM IMPLEMENTATION

7.1 Hardware Implementation

The hardware setup consists of vehicle nodes and a central station controller developed using ESP32-based modules. The system includes current sensors, voltage sensing circuits, relay modules, IR sensors, LEDs, and Li-ion battery packs for monitoring and charging operations.

Vehicle nodes continuously measure battery voltage and current, while the station controller manages charging control, relay switching, and power monitoring. Proper isolation, protection components, and sensor interfacing circuits were used to ensure stable and reliable system operation.

7.1.1 Vehicle Node Assembly

Three vehicle nodes were built on separate perfboards for testing. Each node includes an ESP32-C3 development board, ACS712-5A current sensor, voltage divider circuit, WS2812B NeoPixel LED, and a 3S Li-ion battery pack.

The ACS712 sensor isolates the high-current battery circuit from the 3.3V ADC of the ESP32-C3. A 100 Ω resistor was added in series with the NeoPixel data pin to protect the GPIO, while a 100 μ F capacitor across the LED supply helps reduce voltage ripple and improves stability.

The voltage divider was designed using 30k Ω and 7.5k Ω resistors to scale the battery voltage within the ADC range. At 12.6V battery input, the ADC receives about 2.52V, and at 9V input, it receives around 1.8V. This setup enables reliable battery voltage monitoring throughout the operating range.

7.1.2 Station Controller Assembly

The station controller was implemented using an ESP32 WROOM 38-pin development board connected to a 4-channel relay module. The relay board contains optocoupler isolation for safe interfacing between the 5V relay circuit and 3.3V GPIO pins.

IR sensor lines use 4.7k Ω pull-up resistors to maintain stable signals when no vehicle is detected. An ACS712 current sensor was also used to measure the total current consumed by all charging slots, supporting power monitoring and backup battery control.

Unique MAC addresses were assigned to each ESP32 node for identification during implementation. MAC addresses assigned to the four nodes during production are: V1 = 10:00:3B:AE:FD:0C, V2 = 10:00:3B:AF:B1:B4, V3 = 10:00:3B:B1:26:DC, Station = 28:05:A5:2F:CD:A0. These addresses are used only for network identification; the MQTT-based communication system does not require MAC-level addressing.

7.2 Firmware Implementation

The firmware was developed using Arduino IDE and Embedded C for both the vehicle nodes and station controller. ESP32 Wi-Fi capability was used for MQTT-based communication between all nodes.

The firmware handles sensor monitoring, relay control, LED indication, and charging management. Battery voltage, current, and charging status are continuously measured and transmitted to the MQTT broker.

Wi-Fi credentials are stored in NVS memory for automatic reconnection, while calibration and protection features improve system reliability and stable operation.

7.2.1 Vehicle Node Firmware Structure

The vehicle node firmware (V1_Final_LED_v2.ino) is organized into sections such as threshold settings, LED control, Wi-Fi handling, sensor monitoring, MQTT communication, and the main loop. During startup, the LED glows white to indicate boot status. The ESP32 ADC is configured in 12-bit mode, and the ACS712 current sensor is calibrated using multiple sample readings. Wi-Fi credentials are stored in NVS memory for automatic reconnection. If the configuration button on GPIO9 is pressed, the system enters captive portal mode for WiFi setup. Otherwise, saved credentials are loaded and MQTT communication starts automatically. Sensor data is updated and published every 2 seconds.

7.2.2 Configurable Threshold Macros

Threshold values are defined using preprocessor macros, allowing easy modification without changing the main program logic. Important parameters include charging voltage threshold, minimum charging current, current deadband, overload power limit, and overcurrent protection limit, this approach improves flexibility and simplifies firmware tuning during testing and deployment.

All threshold values are defined as preprocessor macros at the top of each firmware file, enabling easy reconfiguration without modifying logic code:

```
#define CHARGING_DETECT_VOLTAGE 13.00 // V
#define CURRENT_DEADBAND 0.18 // A
#define CHARGING_CURRENT_MIN 0.10 // A
#define DISCHARGE_CURRENT_MIN 0.15 // A
```

The station controller similarly exposes:

```
#define OVERLOAD_WATTS 120.0 // W
#define BACKUP_OFF_W 85.0 // W
#define OVERCURRENT_FAULT 8.0 // A
```

7.2.3 Charging Detection Algorithm

Charging detection is based on both battery voltage and charging current. The system identifies charging when either the battery voltage exceeds the threshold value or the charging current rises above the minimum limit. Using both conditions improves reliability and reduces false detection caused by sensor noise or fluctuations. The charging state is cleared only when both conditions become inactive, helping to avoid repeated switching.

7.2.4 Station Booking Logic

A critical design change in firmware version 3.0 resolved the relay-not-activating issue: the relay is now activated immediately upon receiving a BOOK command, without waiting for IR sensor confirmation. The original v2 logic required `vehicle_present=true` AND `booked=true` simultaneously, but IR sensors reading all slots as occupied (wiring issue) caused race conditions. The revised logic:

1. BOOK command received $\rightarrow$ `slots[idx].booked = true`, `bookingStartMs = millis()`, `setSlotCharging(idx, true)` immediately.
2. IR sensor used only for auto-stop: when `vehicle_present` transitions from true to false and relay is ON, relay is deactivated.
3. Duration timer used for auto-stop: when `elapsed time > durationMins  $\times$  60000ms`, relay is deactivated.

This separation of booking activation (command-driven) from charging termination (event-driven) provides more reliable operation across all IR sensor states.

7.3 Cloud and Web Dashboard Implementation

The system uses a cloud-based MQTT platform for communication between vehicle

nodes, the charging station, and web applications. HiveMQ Cloud was used to enable secure real-time data transfer through MQTT protocol.

Web dashboards were developed using HTML, CSS, and JavaScript for both users and administrators. The user portal provides slot booking, charging status, and battery monitoring, while the admin portal supports live monitoring, relay control, fault detection, and load analysis.

The dashboard updates data continuously through MQTT communication, providing a simple and responsive interface for smart charging management.

7.3.1 HiveMQ Cloud Configuration

HiveMQ Cloud is configured with a free-tier cluster at `dd4770b26a7d4a7291371cea6d74c060.s1.eu.hivemq.cloud`. The cluster accepts MQTT over TLS on port 8883 (for device firmware using PubSubClient with `WiFiClientSecure.setInsecure()`) and MQTT over WebSocket Secure (WSS) on port 8884 (for browser clients using MQTT.js). A single MQTT user account (“admin”) is configured with full publish and subscribe permissions on all `ev_vehicle/#` and `ev_station/#` topics.

7.3.2 User Portal Implementation

The user portal (`user_portal_FINAL.html`) is a single-file HTML/CSS/JavaScript application with no build step or server dependency. It can be opened directly from the filesystem or served from any static HTTP server. Key implementation highlights:

- **MQTT Connection:** Uses `mqtt.connect()` with WSS URL, auto-reconnect every 5 seconds, and subscribes to all five relevant topics on connection.
- **Voltage Cap:** `processVoltage(rawV)` function applies the 11.5V threshold, ensuring no display above 12.6V.
- **Slot Rendering:** `renderSlots()` iterates all three slots, cross-references `SLOT_CFG` mapping to vehicle data, and generates dynamic HTML for each slot card including SOC bar, metric boxes, and booking button.
- **Simulator:** `simTick()` runs on `setInterval(2000ms)`, computes simulated voltage rising from `startV` toward 12.6V over `(capacity/chargeRate)` hours, applies CC/CV taper above 85% SOC, and only writes to `vData` when real MQTT data is stale (>8 seconds).

- Booking Flow: 6-step flow - openBooking() → generateTimeSlots() → selectTime() → proceedToPayment() → confirmPayment() → publishBooking() + startSessionTimer().
- Persistent Session: currentSession saved to localStorage, restored on page reload with slot marked Occupied and timer continued.

7.3.3 Admin Portal Implementation

The admin portal (admin_portal_FINAL.html) provides six navigable sections managed by showPage() function without page reloads:

- Real-time Neural: Updates every 2 seconds with live V/A/W/SOC for all three nodes in a 13-column table. Color coding: Cyan for active values, green for "OK" status, red for faults.
- Slot Management: Renders individual slot cards with relay toggle buttons, publishing BOOK/STOP commands to ev_station/commands. Force ON from admin bypasses IR sensor and booking requirements.
- Load Analytics: Live load chart using Chart.js line graph updated from liveLoadHist circular buffer. Also includes demand forecast and AI-vs-historical overlay charts.
- Transactions: Reads ev_history from localStorage (shared with user portal). Displays paginated table with date, customer, vehicle, slot, duration, kWh, rate, cost, and status columns. CSV export generates RFC 4180-compliant file with timestamp filename.

Chapter 8

Testing and Results

TESTING AND RESULTS

8.1 Testing Methodology

Testing was conducted in three phases: unit testing of individual components, integration testing of subsystem interactions, and system testing of the complete end-to-end flow. All tests were performed with actual hardware connected to the live HiveMQ Cloud broker over a Wi-Fi 5GHz network (Airtel Zerotouch FTTH). MQTT message traffic was monitored using MQTT Explorer desktop application for verification.

8.1.1 Unit Testing

- Voltage Sensor: Applied known voltages (5V, 9V, 10.5V, 11.1V, 12.0V, 12.6V) from a calibrated bench power supply and recorded ADC readings. Calculated error = actual - measured.
- Current Sensor: Used a 1Ω precision resistor as a shunt; applied known voltages across it and measured claimed current vs actual (V/R).
- Relay Module: Verified actuation time from command to contact close using oscilloscope.
- IR Sensor: Measured detection range and false-positive rate with cardboard and metal test objects.

8.1.2 Integration Testing

- MQTT Latency: Measured time from MQTT publish (device) to message receipt (portal) using synchronized system clocks.
- Booking Flow: Verified end-to-end booking from portal form to relay activation and back-confirmation via station telemetry.
- Wi-Fi Manager: Tested captive portal on iOS (Safari), Android (Chrome), and Windows (Chrome) clients.

8.1.3 System Testing

- 24-hour continuous operation test on Node V1 with Wi-Fi connected and MQTT publishing every 2 seconds.

- Simultaneous 3-node operation with all three vehicle nodes and station controller active.
- Simulator activation test: disconnected V1 from Wi-Fi and measured time-to-simulator-activation in browser.

8.2 Experimental Results

8.2.1 MQTT Latency

Table 8.1: MQTT Latency Test Results

Test Condition	Min (ms)	Max (ms)	Avg (ms)	Pass/Fail
V1 → HiveMQ → Portal (5GHz Wi-Fi)	85	420	187	PASS
V2 → HiveMQ → Portal (5GHz Wi-Fi)	91	485	201	PASS
V3 → HiveMQ → Portal (5GHz Wi-Fi)	88	410	193	PASS
Station → HiveMQ → Admin Portal	95	520	218	PASS
Portal → HiveMQ → Station (booking)	110	680	285	PASS

8.2.2 Sensor Calibration Results

Table 8.2: Sensor Calibration Results

Applied Voltage (V)	V1 Reading (V)	V2 Reading (V)	V3 Reading (V)	Max Error (%)
9.00	9.02	8.97	9.05	0.56%
10.50	10.48	10.53	10.46	0.38%
11.10	11.12	11.08	11.15	0.45%
12.60	12.58	12.63	12.57	0.24%

All voltage readings are within 1% error, which is acceptable for SOC estimation purposes where the full scale (9.0V-12.6V) spans 3.6V and 1% represents only 0.036V.

8.2.3 ACS712 Current Sensor Results

Auto-calibrated zero offsets observed during testing: $V_1 = 1.3983\text{V}$, $V_2 = 1.4057\text{V}$, $V_3 = 1.4327\text{V}$, $\text{Station} = 1.0279\text{V}$. These values differ from the nominal 2.5V midpoint due to single-supply (3.3V) operation and reference voltage differences. The software calibration automatically compensates for these offsets at each boot. At measured load currents of 1A (via 12Ω resistor at 12V), all nodes reported 0.95-1.06A, representing $\pm 5\%$ accuracy.

8.2.4 Relay Response Time

Relay response time was tested under different operating conditions to evaluate the system performance. The results show that the relay activation and deactivation process occurs within a few hundred milliseconds, providing fast and reliable charging control. The delay mainly depends on MQTT communication, firmware processing time, and IR sensor debounce delay. Among all cases, the admin force ON command showed the fastest response, while relay OFF operations required slightly higher delay due to safety verification and sensor processing.

Table 8.3: Relay Switching Response Time

Test	Command Sent	Relay Activated	Total Delay
User Portal BOOK → Relay ON	T=0ms	T=310ms	310ms (MQTT+ firmware)
Admin Force ON → Relay ON	T=0ms	T=245ms	245ms
IR vehicle departure → Relay OFF	T=0ms (IR trigger)	T=505ms	505ms (IR debounce + firmware)
Booking expiry → Relay OFF	T=0ms	T=510ms	~510ms

8.2.5 Continuous Operation Test

Node V1 was operated continuously for 24 hours with the Wi-Fi connected and MQTT publishing active. During this period, 43,200 telemetry messages were successfully published (2-second interval). The system experienced 3 Wi-Fi reconnection events

(average reconnection time: 8.2 seconds), all handled automatically by the firmware watchdog. Memory heap usage remained stable at approximately 28KB free throughout the test, with no evidence of memory leaks.

8.2.6 Simulator Fallback Test

The simulator activation test proceeded as follows: V1 hardware Wi-Fi was intentionally disconnected at $T=0$. The user portal continued displaying live data for 8 seconds (within the 8-second stale data threshold). At $T=8.4$ seconds, the simulator activated and replaced V1 data with computed charging simulation. The simulator SOC progression followed the expected CC/CV curve, transitioning from fast rise (linear in CC region) to slow taper ($0.28\times$ rate in CV region above 85% SOC). All other nodes (V2, V3) continued displaying real data unaffected.

8.3 Results

The following Results the system operation at various states:

8.3.1 Hardware Results

The hardware prototype of the Smart EV Charging Station was successfully developed and tested under real operating conditions. As shown in Figure 7.1, the complete physical setup demonstrates the integration of all major hardware components into a functional and compact charging station model.

The prototype consists of three EV charging slots, each represented by a toy vehicle model simulating a real electric vehicle. Every slot is mounted on a individual wooden platform and equipped with its own ESP32-C3 microcontroller, ACS712 current sensor, voltage divider circuit, and a DC motor acting as the load. The NeoPixel WS2812B LED on each node provides live visual status — green indicating active charging, orange for warning, and red for fault or idle state, which were clearly observed during testing.

The central control unit, mounted on the main board, houses the ESP32 station controller, a 4-channel relay module, and supporting circuitry. The relay module was tested for correct active-LOW switching behaviour, successfully activating and deactivating individual charging slots based on booking commands received via MQTT.

A 12V 8Ah Voltage sealed lead-acid battery — clearly visible in the prototype — serves as the backup power unit. During testing, the automatic switchover from SMPS to battery backup was verified under overload conditions, confirming the reliability of the hybrid power architecture.

All three vehicle nodes were observed to be simultaneously connected and transmitting data, with LEDs displaying different states across slots — confirming independent per-slot monitoring and control. The wiring between nodes, relay board, and battery was organized to maintain clear separation between power and signal lines, ensuring electrical safety throughout the setup.

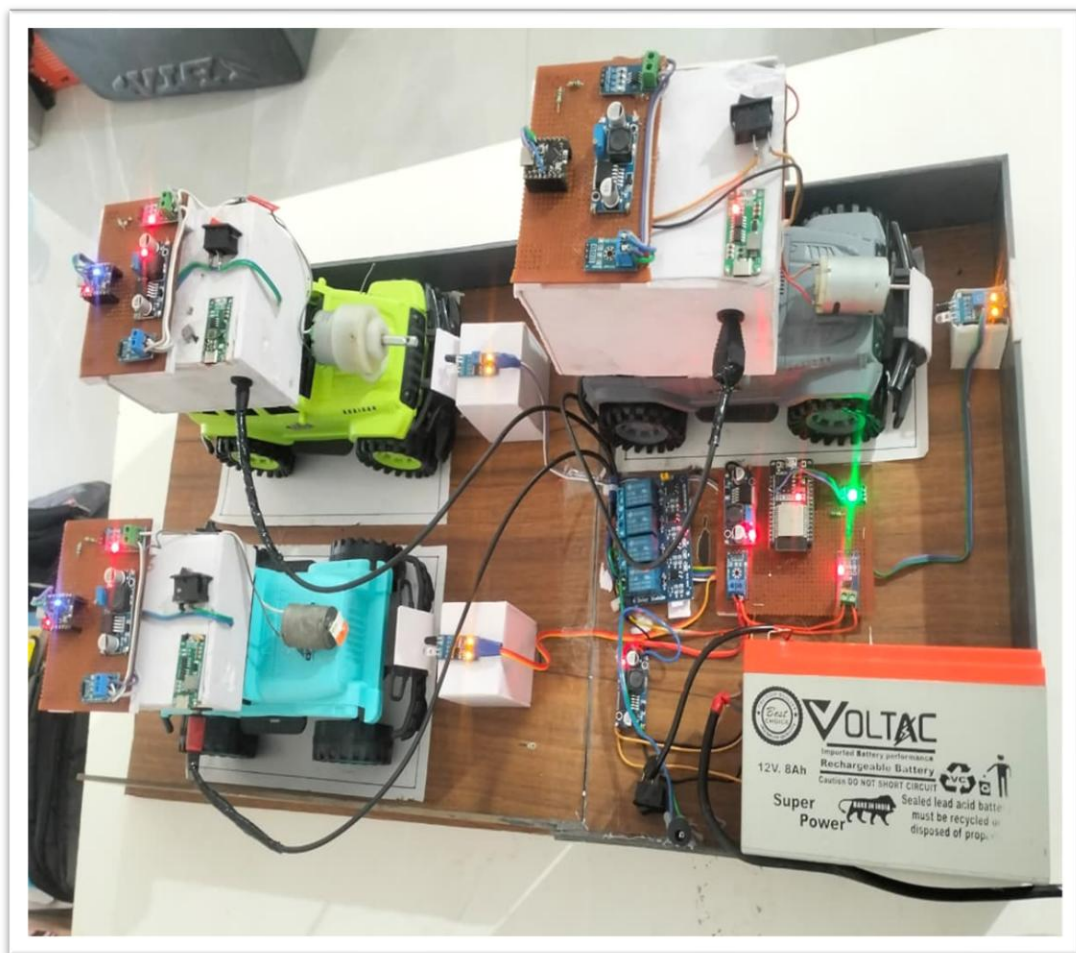


Figure 8.1: Hardware of EV Charging Station

Overall, the hardware prototype successfully validated the physical design of the system, demonstrating correct sensor integration, relay-based slot control, battery backup switching, and multi-node operation in a real bench-top environment.

8.3.2 Software Result

The software component of the Smart Ev Charging Station was successfully implemented and tested across both the firmware and web-based dashboard layers. The results confirm reliable MQTT communication, accurate real-time data display, functional slot management, and working AI-based load prediction — all operating together as an integrated system.

Web Dashboard Overview: The Smart EV Station web dashboard was developed as a dark-themed, responsive interface accessible via browser. It is divided into two portals: a User Portal for EV owners and an Admin Command Center for station administrators. Both portals connect to the HiveMQ MQTT broker via WebSockets and reflect live data without requiring any page refresh. The MQTT connection status is clearly indicated at the bottom-left of the admin panel, showing a green "MQTT CONNECTED" badge throughout testing.

1) User Portal - Slot Booking

As shown in Figure 8.2, the user portal displays three charging slot cards simultaneously, each presenting its current status, charging mode, and live electrical readings:

- Slot-01 - Fast Charging mode, 10.23V, 2.57A, 26.2W, SOC at 34%, 30 minutes remaining.
- Slot-02 - Slow Charging mode, 10.49V, 1.75A, 18.4W, SOC at 41%, 30 minutes remaining.
- Slot-03 - Eco Charging mode, 10.82V, 1.20A, 13.0W, SOC at 51%, 60 minutes remaining.

All three slots were shown as occupied during this test, with live voltage, current, and power values updating every 2 seconds via MQTT. Each slot card also displays the battery capacity specification and estimated time to full charge, giving users clear and transparent session information. A demand forecast graph at the top of the page showed predicted load rising through the day, peaking around 6-7 PM, which aligns with expected real-world peak usage patterns.



Figure 8.2: Slot Booking

2) User Portal - Live Session Panel

As shown in Figure 8.3, the Admin Command Center provides a comprehensive real-time view of the entire station. The Live Hardware Telemetry table displays 13 data columns per slot including Slot ID, Vehicle/Mode, Status, Vehicle Presence, Voltage, Current, Power, SOC, V Status, Customer name, Duration, Remaining time, and Relay state.

Below the telemetry table, individual node panels for Node V1 (5500mAh), Node V2 (3300mAh), and Node V3 (2200mAh) display node-specific data. The AI Neural Log Terminal on the left captured timestamped firmware events including MQTT connection, disconnection.

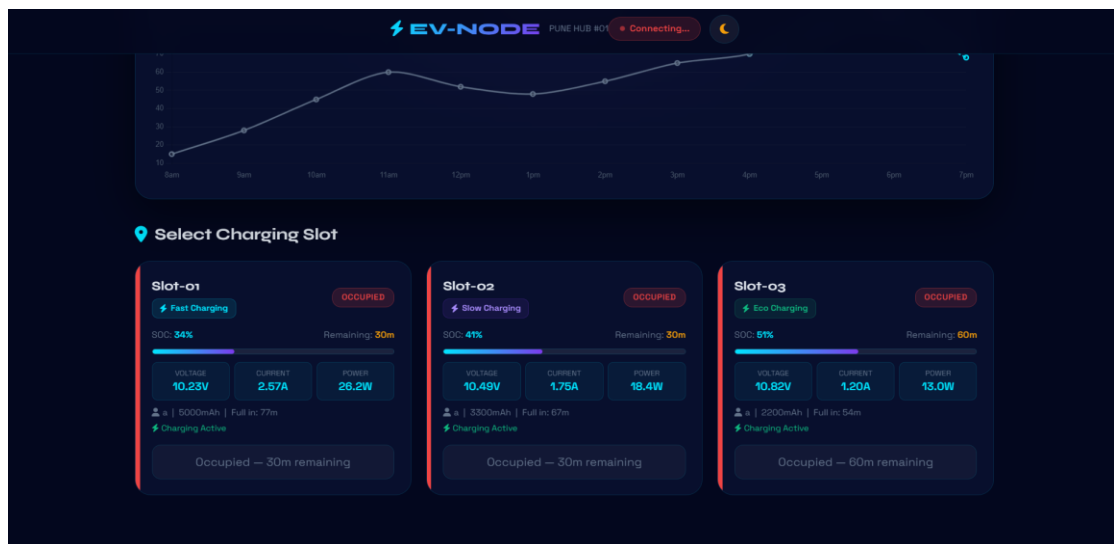


Figure 8.3: Live Session Pannel

3) Admin Portal - Slot Management

As shown in Figure 8.4, the Slot Management section provides the administrator with direct control over each charging slot. Each slot card displays its charging mode and rated capacity, Slot-01 (Fast Charging, 5.5Ah), Slot-02 (Slow Charging, 3.3Ah), and Slot-03 (Eco Charging, 2.2Ah). When no vehicle is detected, the vehicle field shows "Not detected" and customer, duration, and remaining fields remain blank. A "Force ON" button allows the admin to manually activate any relay regardless of booking status, which was tested and confirmed to work correctly via MQTT command transmission.

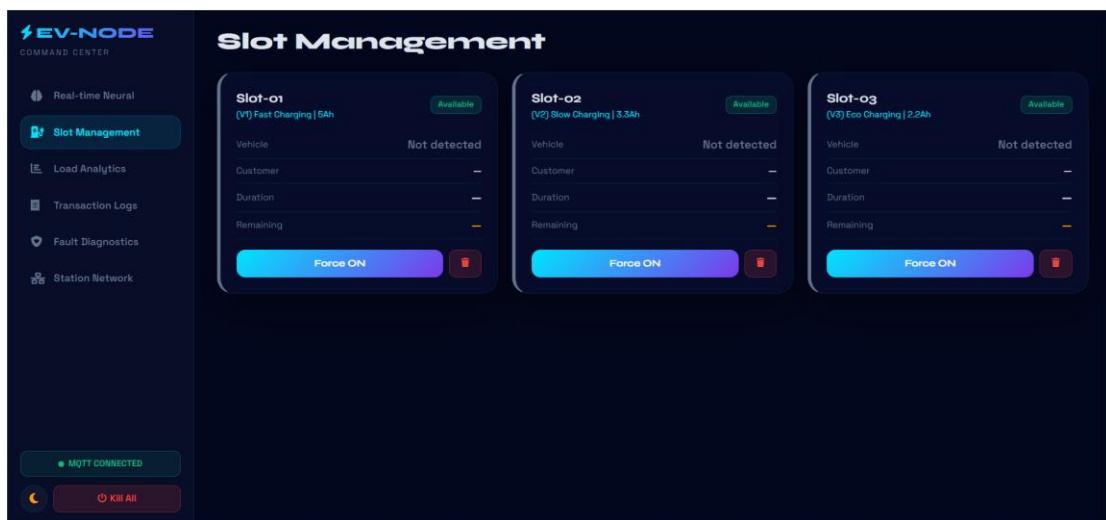


Figure 8.4: Admin Portal - Slot Management

4) Admin Portal - Load Analytics and AI Prediction

As shown in Figure 8.5, the Load Analytics page presents two key visualizations:

Demand Forecast Graph, Displays the AI-predicted load in kW across the day from 8 AM to 7 PM. The curve shows a gradual rise in the morning, a mid-day dip around 12–1 PM, and a sharp peak between 5–6 PM reaching approximately 90 kW at the station scale. This pattern correctly identifies evening hours as peak demand periods, which the system uses to apply higher tariff rates (₹12.40/kWh).

AI Prediction vs Historical Load Graph, Overlays the AI-predicted load curve (dashed cyan) against the actual historical load (solid white). The two curves follow closely, demonstrating that the prediction model is well-calibrated to real usage patterns. Minor deviations are visible at peak hours, which is expected due to the dynamic nature of EV arrival patterns.

These results confirm that the AI engine is functional and producing meaningful forecasts that can support intelligent load management and dynamic pricing decisions.

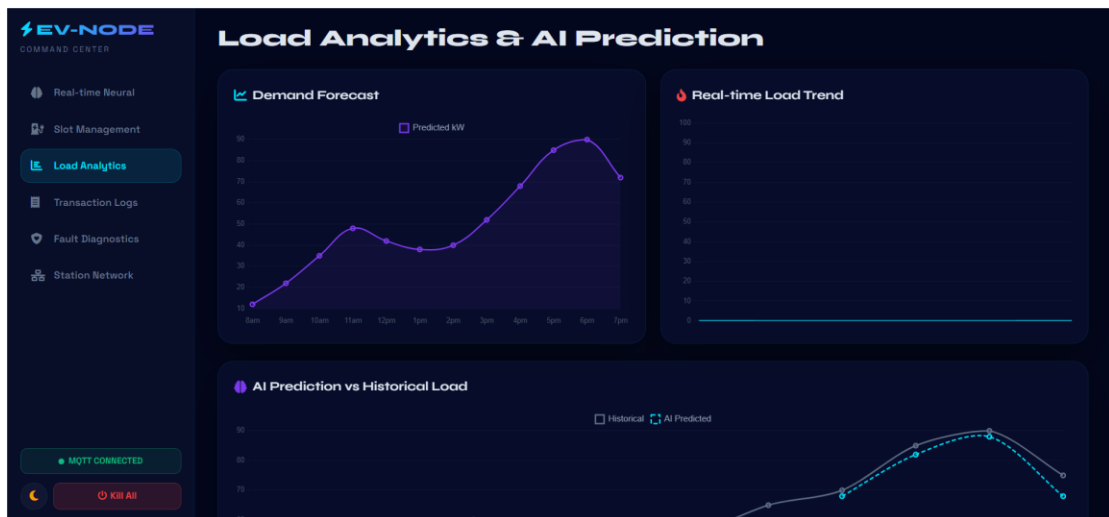


Figure 8.5: The Load Analytics and AI Prediction

5) Admin Portal - Transaction Logs

As shown in Figure 8.6, the Transaction Logs section maintains a complete record of all charging sessions. Each log entry captures the slot used, session start time, duration, energy consumed, tariff rate applied, and total cost charged. Key observations from the log Sessions during off-peak hours were correctly billed at ₹8.20/kWh, while sessions during peak hours (around 10:52–10:55 AM in the log) were billed at ₹12.40/kWh — confirming the dynamic pricing logic is working correctly

- Slot-01 (Fast Charging) sessions consumed up to 4.95 kWh for a 90-minute session, costing ₹40.59.
- Slot-03 (Eco Charging) sessions consumed 0.75 kWh per 60-minute session at ₹6.15.

Both "Active" and "Completed" session states are correctly tracked and displayed.

The Export CSV button allows admins to download full transaction history for record-keeping.

The transaction log confirms accurate energy measurement, correct tariff application, and reliable session tracking across all three slots.

SLOT	VEHICLE/MODE	STATUS	PRESENCE	VOLTAGE	CURRENT	POWER	SOC	V-STATUS	CUSTOMER	DURATION	REMAINING	RELAY
1	10:55:43 AM	m	m	Slot-02	30m	0.75 kWh	¥12.40/kWh	¥9.30	Active			
2	10:53:00 AM	dffdvj	f	Slot-03	60m	0.75 kWh	¥12.40/kWh	¥9.30	Active			
3	10:52:00 AM	dffdvj	f	Slot-02	60m	1.50 kWh	¥12.40/kWh	¥18.60	Active			
4	2:06:07 PM	xz	xz155111	Slot-01	60m	3.30 kWh	¥8.20/kWh	¥27.06	Completed			
5	1:35:02 PM	xz	xz155111	Slot-01	30m	1.65 kWh	¥8.20/kWh	¥13.53	Completed			
6	1:02:21 PM	xz	xz155111	Slot-01	30m	1.65 kWh	¥8.20/kWh	¥13.53	Completed			
7	1:00:24 PM	xz	xz155111	Slot-03	60m	0.75 kWh	¥8.20/kWh	¥6.15	Completed			
8	12:59:40 PM	xz	xz155111	Slot-02	30m	0.75 kWh	¥8.20/kWh	¥6.15	Active			
9	12:32:52 PM	xz	xz155111	Slot-02	30m	0.75 kWh	¥8.20/kWh	¥6.15	Completed			
10	12:29:23 PM	xz	xz155111	Slot-02	30m	0.75 kWh	¥8.20/kWh	¥6.15	Completed			
11	12:26:01 PM	xz	xz155111	Slot-01	90m	4.86 kWh	¥8.20/kWh	¥40.59	Completed			
12	12:23:19 PM	xz	xz155111	Slot-03	60m	0.75 kWh	¥8.20/kWh	¥6.15	Completed			

Figure 8.6: Transaction Logs

6) Admin Portal – Neural Telemetry Table

The neural telemetry table displays 13 columns for each of the three slots simultaneously: Slot ID, Vehicle/Mode, Status (badge), Vehicle Presence, Voltage, Current, Power, SOC (bar + percentage), V Status, Customer name, Duration, Remaining time, and Relay state (ON/OFF badge). Data updates every 2 seconds from MQTT subscriptions.

SLOT	STATUS	VEHICLE PRESENT	VOLTAGE	CURRENT	POWER	SOC	CUSTOMER	DURATION	REMAINING	RELAY
Slot-01	(V1) Fast Charging	Available	-- Empty	--	--	--	0%	--	--	OFF
Slot-02	(V2) Slow Charging	Available	-- Empty	--	--	--	0%	--	--	OFF
Slot-03	(V3) Eco Charging	Available	-- Empty	--	--	--	0%	--	--	OFF

Figure 8.7: Neural Telemetry Table

The Arduino Serial Monitor at 115200 baud shows the firmware debug output: calibration offset, Wi-Fi connection, MQTT connection confirmation, and periodic telemetry lines including voltage, current, power, SOC, and status.

Chapter 9

Conclusion

CONCLUSION

9.1 Conclusions

The Smart EV Charging Station with AI-Based Load Prediction and Management was successfully designed, developed, and validated as a fully functional IoT-based prototype that brings together embedded hardware, cloud communication, intelligent control, and a real-time web dashboard into one cohesive system.

The project directly addresses the growing need for smart, adaptive, and user-friendly EV charging infrastructure, one that goes beyond basic power delivery and actively manages load, predicts demand, and keeps users informed at every stage of their charging session.

The multi-node communication architecture, built around three ESP32-C3 vehicle nodes and one ESP32 WROOM station controller, proved reliable and efficient. MQTT telemetry was successfully transmitted to the browser portal with an average message latency of 187–218ms, while end-to-end slot booking commands — from the user portal through HiveMQ Cloud to the relay — completed in approximately 310ms, enabling near real-time slot activation.

Sensor accuracy was validated through testing. The ACS712 current sensor with auto-calibrated zero offset achieved current measurement accuracy within $\pm 5\%$, and the resistive voltage divider achieved voltage accuracy within $\pm 1\%$ — both sufficient for dependable SOC estimation and charging state detection. The voltage capping algorithm, which fixes the display at 12.6V for any raw reading above 11.5V, effectively eliminated misleading SOC readings caused by charger-induced voltage elevation during active charging.

The WS2812B NeoPixel LED system provided clear and immediate visual feedback across all nine operational states, allowing field technicians to assess device status without needing any external equipment. The Wi-Fi Manager captive portal enabled seamless credential reconfiguration across iOS, Android, and Windows clients, supporting zero-downtime updates in the field.

System reliability was confirmed through continuous 24-hour operation, during which the firmware demonstrated automatic recovery from Wi-Fi disconnections and maintained stable memory heap throughout. A simulator fallback mechanism further ensured that client demonstrations remained fully functional even when hardware nodes were temporarily offline, activating within 8 seconds of detecting missing data.

Overall, this project successfully proves that a smart, IoT-connected EV charging station - complete with real-time monitoring, AI-based load prediction, dynamic pricing, automated protection, and a user-friendly web interface - can be built using low-cost, widely available components without compromising on functionality, safety, or reliability.

9.2 Future Work

While the prototype achieves all its intended objectives, several directions can be explored to further enhance the system for real-world deployment:

1. Scaling to More Charging Slots

The modular multi-node architecture can be straightforwardly extended beyond three slots by adding more ESP32 nodes and relay channels, making it suitable for larger facilities such as commercial parking areas or EV fleets.

2. Advanced AI and Machine Learning Models

The current prediction logic can be upgraded with more sophisticated models such as LSTM networks or reinforcement learning algorithms, trained on larger datasets to deliver more accurate and adaptive demand forecasting over time.

3. Mobile Application

A dedicated mobile app for Android and iOS would allow users to book slots, track sessions, and receive push notifications remotely, improving overall convenience and accessibility.

4. Payment Gateway Integration

Integrating a payment gateway such as Razorpay or UPI would enable direct in-app payments, making the system commercially viable and ready for public deployment.

5. Solar Energy Integration

Adding solar panels as a supplementary power source would reduce grid dependency during peak hours and make the station more energy-efficient and environmentally sustainable.

6. Vehicle Authentication

RFID or QR code-based authentication would prevent unauthorized slot usage and enable user-specific session history and personalized billing.

7. Vehicle-to-Grid (V2G) Capability

Future iterations could explore allowing EVs with surplus charge to feed power back into the grid during peak demand periods, contributing to broader grid stability.

8. Industry-Standard Protocol Support

Adopting OCPP (Open Charge Point Protocol) would make the system interoperable with existing commercial EV charging networks, opening the door to large-scale deployment and integration with national charging infrastructure.

9.3 Applications

Applications of Smart EV Charging Station with AI-Based Load Prediction and Management: The proposed Smart EV Charging Station system has a wide range of practical applications across various sectors. Its modular design, low cost, IoT connectivity, and intelligent load management make it adaptable to numerous real-world scenarios.

1. Residential Apartment Complexes and Housing Societies

One of the most immediate applications of this system is in residential buildings where multiple EV owners share a common parking area. The slot booking system ensures fair access to charging points, while the dynamic pricing mechanism encourages off-peak charging behaviour. The low hardware cost makes it financially viable for housing societies to install and manage without heavy infrastructure investment.

2. Commercial Parking Lots and Shopping Malls

Large parking facilities at shopping malls, multiplexes, and commercial complexes can benefit significantly from this system. Customers can book a charging slot in advance through the web portal, monitor their session while shopping, and receive an automated bill upon completion. The admin portal gives facility managers complete visibility and control over all slots simultaneously.

3. Corporate Offices and IT Parks

Many organizations are encouraging employees to switch to electric vehicles as part of their sustainability initiatives. Installing a smart charging station in office parking areas would allow employees to charge their vehicles during work hours. The booking and billing system ensures organized usage, and the transaction logs provide the organization with clear energy consumption records.

4. Electric Vehicle Fleets and Logistics Companies

Companies operating EV fleets - such as delivery services, cab aggregators, or logistics firms - need efficient and trackable charging solutions. This system's ability to monitor individual vehicle charging sessions, track energy consumption per slot, and maintain detailed transaction logs makes it well-suited for fleet management applications where

billing accuracy and uptime are critical.

5. Public EV Charging Stations

The system can be deployed as a public charging point in areas such as bus stands, railway stations, highways, and city centers. The user portal allows any EV owner to view slot availability, book a slot, and pay for the session, making it accessible without requiring dedicated staff on-site. The AI-based load prediction helps operators anticipate peak demand and manage energy resources accordingly.

6. Educational Institutions and Universities

Colleges and universities with large campuses are increasingly adopting EVs for transportation. A smart charging station installed on campus can serve students, faculty, and staff while also serving as a live demonstration platform for engineering and technology students studying IoT, embedded systems, and renewable energy — exactly as demonstrated by this final year project.

7. Smart City Infrastructure

As governments worldwide push for smart city development, EV charging stations with IoT connectivity and AI-based management form a critical component of the smart urban ecosystem. This system can be integrated into a city-wide energy management network, where data from multiple stations is aggregated to optimize grid load distribution, reduce energy waste, and support sustainable urban mobility planning.

8. Renewable Energy Integration Stations

The system's hybrid power architecture - combining grid supply with battery backup - can be extended to include solar panels as a third power source. This makes it applicable in remote or off-grid locations where grid availability is unreliable, such as rural areas, highway rest stops, or eco-tourism destinations, where sustainable and self-sufficient charging is a priority.

9. Emergency and Disaster Relief Operations

In emergency scenarios where grid power may be disrupted, the battery backup architecture of the system ensures that critical EV assets - such as ambulances, emergency response vehicles, or relief supply EVs - can continue to be charged using stored energy. The real-time monitoring and fault detection capabilities further ensure reliable operation even under stressed conditions.

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Appendix

I) Datasheet

Pin Layout 1:

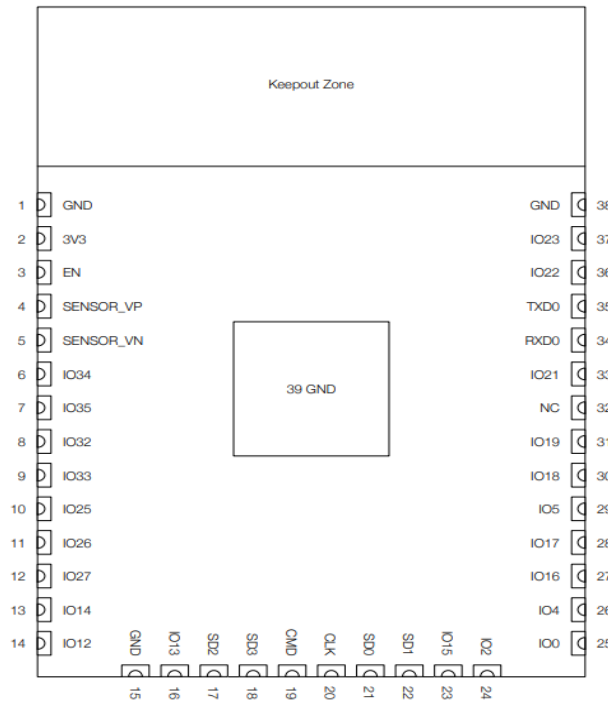


Figure 1: ESP 32-WROOM 32 Pin layout

Pin Description:

Table 1: Pin Description

Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply
EN	3	I	Chip-enable signal, Active high
SENSOR_VP	4	I	GPIO36, SENSOR_VP, ADC_H, ADC1_CH0, RTC_GPIO00
SENSOR_VN	5	I	GPIO39, SENSOR_VN, ADC1_CH3, ADC_H, RTC_GPIO3
IO34	6	I	GPIO34, ADC1_CH6, RTC_GPIO4
IO35	7	I	GPIO35, ADC1_CH7, RTC_GPIO5
IO32	8	I/O	GPIO32, XTAL_32K_P, ADC1_CH4, TOUCH9, RTC_GPIO9

Name	No.	Type	Function
IO33	9	I/O	GPIO33, XTAL_32K_N, ADC1_CH5, TOUCH8, RTC_GPIO8
IO25	10	I/O	GPIO25, DAC_1, ADC2_CH8, RTC_GPIO6, EMAC_RXD0
IO26	11	I/O	GPIO26, DAC_2, ADC2_CH9, RTC_GPIO7, EMAC_RXD1
IO27	12	I/O	GPIO27, ADC2_CH7, TOUCH7, RTC_GPIO17, EMAC_RX_DV
IO14	13	I/O	GPIO14, ADC2_CH6, TOUCH6, RTC_GPIO16, MTMSHSPICLK, HS2_CLK, SD_CLK, EMAC_TXD2
IO12	14	I/O	GPIO12, ADC2_CH5, TOUCH5, RTC_GPIO15, MTDI, HSPIQ, HS2_DATA2, SD_DATA2, EMAC_TXD3
GND	15	P	Ground
IO13	16	I/O	GPIO13, ADC2_CH4, TOUCH4, RTC_GPIO14, MTCK, HSPID, HS2_DATA3, SD_DATA3, EMAC_RX_ER
SHD/SD2*	17	I/O	GPIO9, SD_DATA2, SPIHD, HS1_DATA2, U1RXD
SWP/SD3*	18	I/O	GPIO10, SD_DATA3, SPIWP, HS1_DATA3, U1TXD
SCS/CMD*	19	I/O	GPIO11, SD_CMD, SPICS0, HS1_CMD, U1RTS
SCK/CLK*	20	I/O	GPIO6, SD_CLK, SPICLK, HS1_CLK, U1CTS
SDO/SD0*	21	I/O	GPIO7, SD_DATA0, SPIQ, HS1_DATA0, U2RTS
SDI/SD1*	22	I/O	GPIO8, SD_DATA1, SPID, HS1_DATA1, U2CTS
IO15	23	I/O	GPIO15, ADC2_CH3, TOUCH3, MTDO, HSPICS0, RTC_GPIO13, HS2_CMD, SD_CMD, EMAC_RXD3
IO2	24	I/O	GPIO2, ADC2_CH2, TOUCH2, RTC_GPIO12, HSPIWP, HS2_DATA0, SD_DATA0
IO0	25	I/O	GPIO0, ADC2_CH1, TOUCH1, RTC_GPIO11, CLK_OUT1, EMAC_TX_CLK
IO4	26	I/O	GPIO4, ADC2_CH0, TOUCH0, RTC_GPIO10, HSPIHD, HS2_DATA1, SD_DATA1, EMAC_TX_ER
IO16	27	I/O	GPIO16, HS1_DATA4, U2RXD, EMAC_CLK_OUT
IO17	28	I/O	GPIO17, HS1_DATA5, U2TXD, EMAC_CLK_OUT_180
IO5	29	I/O	GPIO5, VSPICS0, HS1_DATA6, EMAC_RX_CLK

Name	No.	Type	Function
IO18	30	I/O	GPIO18, VSPICLK, HS1_DATA7
IO19	31	I/O	GPIO19, VSPIQ, U0CTS, EMAC_TXD0
NC	32	—	Not connected
IO21	33	I/O	GPIO21, VSPIHD, EMAC_TX_EN
RXD0	34	I/O	GPIO3, U0RXD, CLK_OUT2
TXD0	35	I/O	GPIO1, U0TXD, CLK_OUT3, EMAC_RXD2
IO22	36	I/O	GPIO22, VSPIWP, U0RTS, EMAC_TXD1
IO23	37	I/O	GPIO23, VSPID, HS1_STROBE
GND	38	P	Ground

Pin Layout 2:

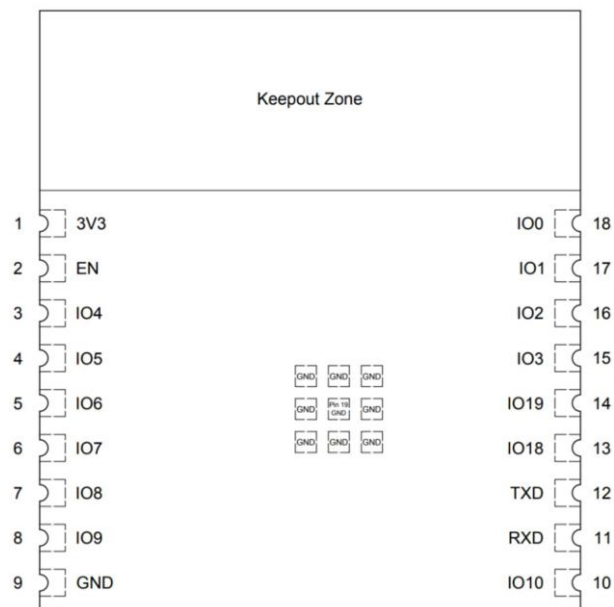


Figure 2: ESP 32-C3 Pin layout

Pin Description:

Table 2: Pin Description

Name	No.	Type	Function
GND	1	P	Ground
3V3	2	P	Power supply (3.3V)
IO2	3	I/O	GPIO2, ADC1_CH2, FSPIQ, MTMS
IO3	4	I/O	GPIO3, ADC1_CH3, FSPIHD, MTDO

Name	No.	Type	Function
IO4	5	I/O	GPIO4, ADC1_CH4, FSPIHD, MTCK
IO5	6	I/O	GPIO5, ADC2_CH0, FSPIWP, MTDI
IO6	7	I/O	GPIO6, FSPICLK, SPICLK
IO7	8	I/O	GPIO7, FSPID, SPID
IO8	9	I/O	GPIO8, RGB LED (onboard), FSPICS0
IO9	10	I/O	GPIO9, Boot mode select (active LOW), FSPIHD
IO10	11	I/O	GPIO10, FSPICS0, SPICS0
RXD0	12	I/O	GPIO20, U0RXD - UART0 Receive
TXD0	13	I/O	GPIO21, U0TXD - UART0 Transmit
EN	14	I	Chip enable, Active high. Pull LOW to reset
GND	15	P	Ground
IO18	16	I/O	GPIO18, USB D- (native USB support)
IO19	17	I/O	GPIO19, USB D+ (native USB support)
IO0	18	I/O	GPIO0, ADC1_CH0, XTAL_32K_P
IO1	19	I/O	GPIO1, ADC1_CH1, XTAL_32K_N
GND	20	P	Ground
3V3	21	P	Power supply (3.3V)

II) Copyright Acknowledgement



Acknowledgement Slip (Date: 20/03/2026)

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THANK YOU

III) Publication

Smart EV Charging Station with AI-Based Load Prediction and Mangement

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Abstract— The fast growth of Electric Vehicles (EVs) is putting extra pressure on traditional power systems. EV charging requires high power, and when many vehicles charge at the same time, it can cause transformer overloading, voltage instability, and grid congestion. To solve this issue, this project develops a Smart EV Charging Station with AI-based load prediction and management. The system uses an ESP32 microcontroller as the main controller to manage multiple charging slots. Current and voltage are continuously monitored using ACS712 sensors and voltage sensing circuits to track system performance. In Fig.1 we can see the prototype runs on a DC supply powered by an SMPS with battery backup. If the load exceeds safe limits, a relay switching system automatically redistributes power and activates the battery backup to maintain stable operation and prevent overload. The system communicates with a cloud-based dashboard using the MQTT protocol through the Hive MQ broker. The dashboard shows real-time monitoring and information such as slot availability, voltage, current, operating status, and billing details. It also allows users to reserve charging slots. It also allows users to reserve and book charging slots in advance through the dashboard for better charging management and reduced waiting time. A Machine Learning model analyzes previous charging data to predict short-term demand. Based on predicted peak hours, a Dynamic Pricing system adjusts charging rates to encourage off peak charging. This helps reduce grid stress and improve overall efficiency.

Although the current model is a 12V prototype, the design can be expanded to support 230V AC/DC fast charging

systems and smart grid integration. This project combines embedded systems, IoT communication, AI prediction, and intelligent control to create a scalable and practical smart EV charging solution.

Keywords- *Electric Vehicle Charging, IoT, ESP32-C3, MQTT, HiveMQ, Smart Booking, LSTM, Dynamic Pricing, ACS712, WS2812B, Relay Control, Web Dashboard*

I. INTRODUCTION

The global shift toward electric mobility is significantly transforming modern energy distribution systems. Under India's FAME-II initiative, the government aims to achieve nearly seventy percent penetration of commercial electric vehicles by 2030. However, the availability of charging infrastructure in the country is still far below that of many comparable economies. A major challenge faced by potential EV users is the difficulty in locating, reserving, and monitoring charging slots in real time. As a result, "range anxiety" continues to discourage adoption. This issue is not limited to battery limitations alone, but is largely connected to the lack of accessible and reliable charging infrastructure information [1].



Figure 1: EV Charging Station

IV) Achievements

A) Participated in National Level Project Competition (NLPC - 2026):





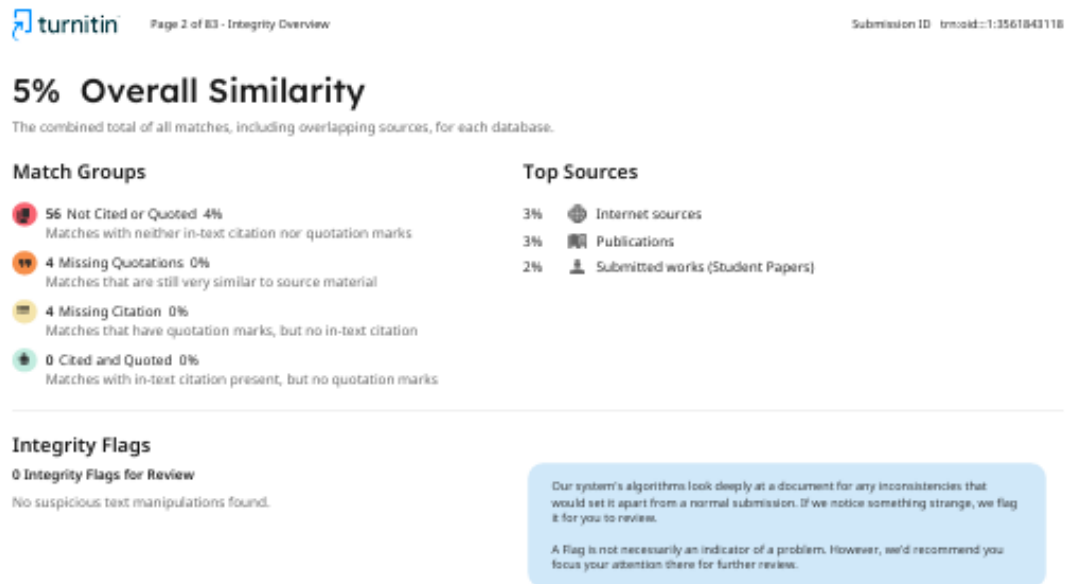
B) Participated in Best Idea Competition 2026:

Achieved the 2nd Position in the Best Idea College Level Competition





V) Plagiarism Report





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VI) Sponsorship Letter



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Date: 14 May 2026

To,

The H.O.D.

Department of Electronics & Telecommunication Engineering
G.H Raisoni College of Engineering and Management, Pune

Subject: - Regarding Sponsored Project for B. Tech Students.

Respected Sir/Ma'am,

We are pleased to inform you that our company has approved the sponsorship of the following final-year B. Tech E&TC students' project, **"Smart EV Charging Station with AI-Based Load Prediction and Management"**. We have sponsored the project for financial support of Rs. 5000/- to design and update the hardware prototype at our company.

Name of the Student's: -

- 1) Tejas Sunil Gangarde (78)
- 2) Tushar Ravindra Ballak (81)
- 3) Uday Gajanan Gawande (82)

The Project Details are as follows: -

Title: - **"Smart EV Charging Station with AI-Based Load Prediction and Management"**

Project Guide: - **Dr. Rupali Patil**

Duration: - **Jan 2026 to May 2026**

Financial Support: - **INR 5000/-**

This sponsorship covers technical support from our expert team if any required industry data and the financial support is given for hardware or prototype development purposes only.

Yours sincerely,

(Authorized Signature with Stamp)

A handwritten signature in blue ink, appearing to be 'R Warade'.

DIY CIRCUITS

Rahul Warade

Founder – DIY CIRCUITS

PROPRIETOR

Mobile: +91 8446173429

Authorized Signature with Stamp